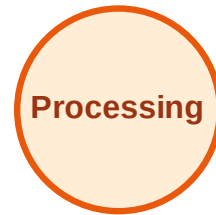


# BFS Traversal

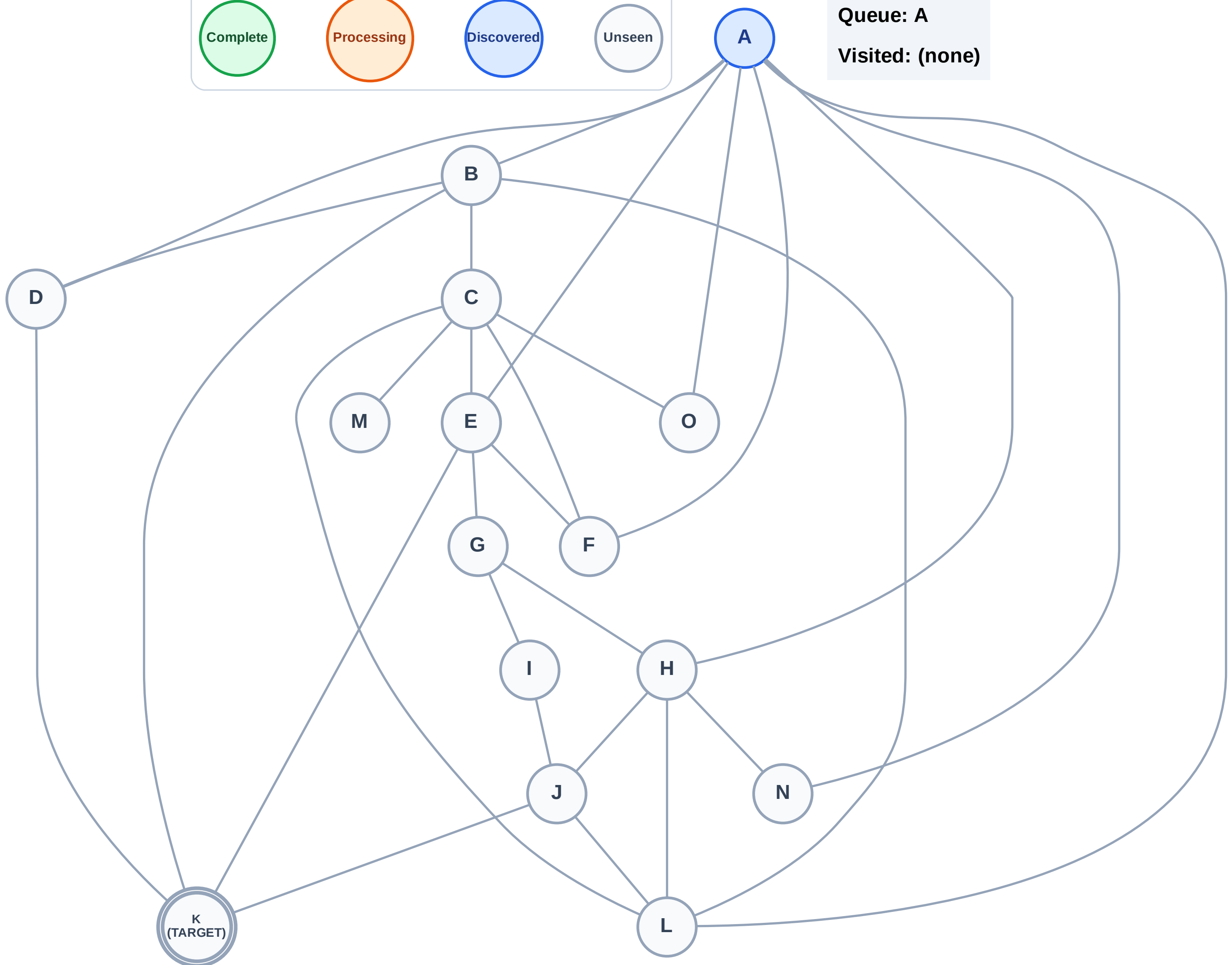
Step 1: Start at A (Seeking target node K)

## Legend



Queue: A

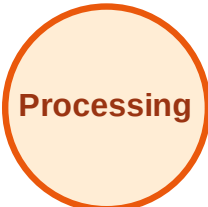
Visited: (none)



# BFS Traversal

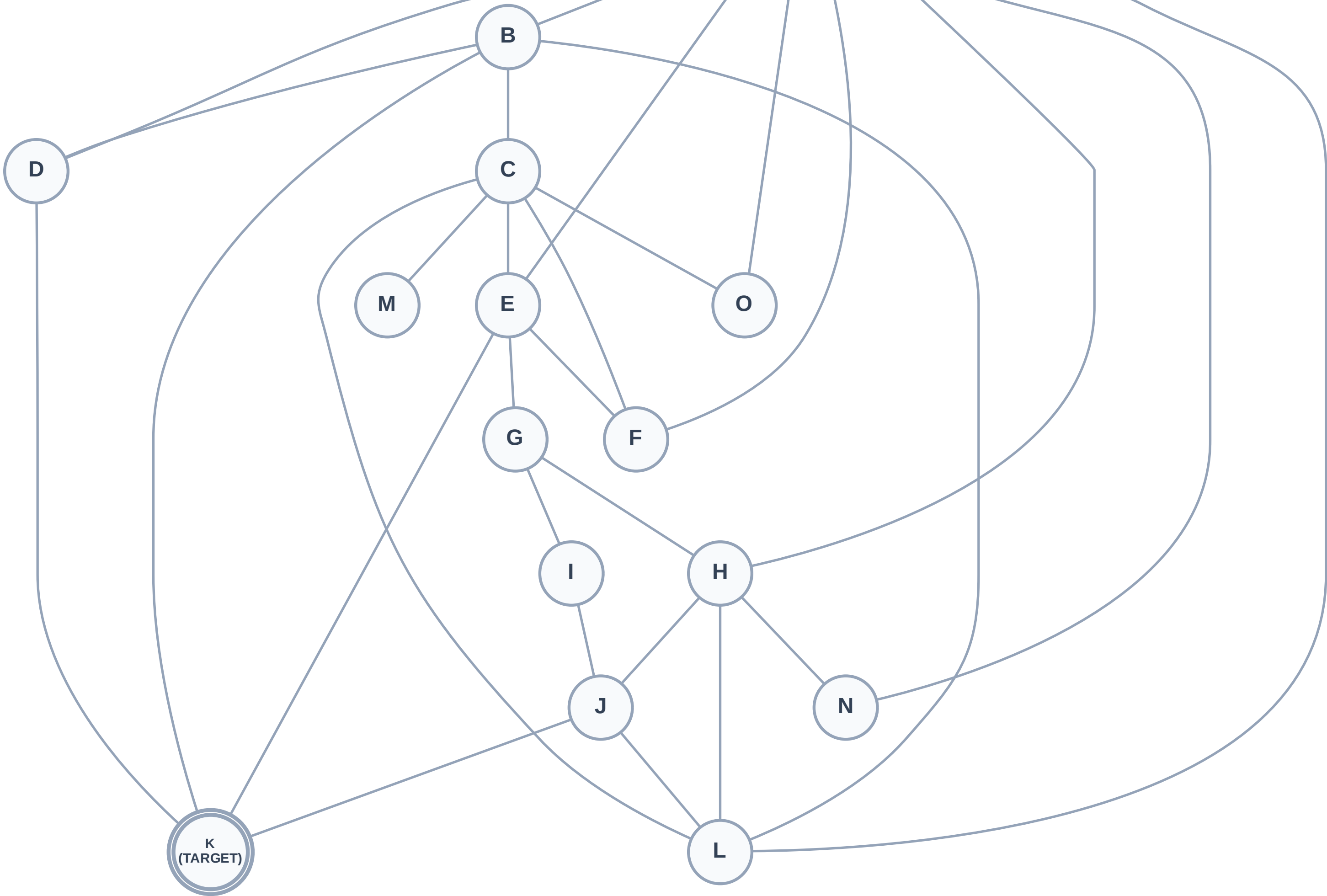
Step 2: Process node A

## Legend



Queue: (empty)

Visited: (none)



# BFS Traversal

Step 3: Discover B, D, E, F, H, L, N, O from A

Legend

Complete

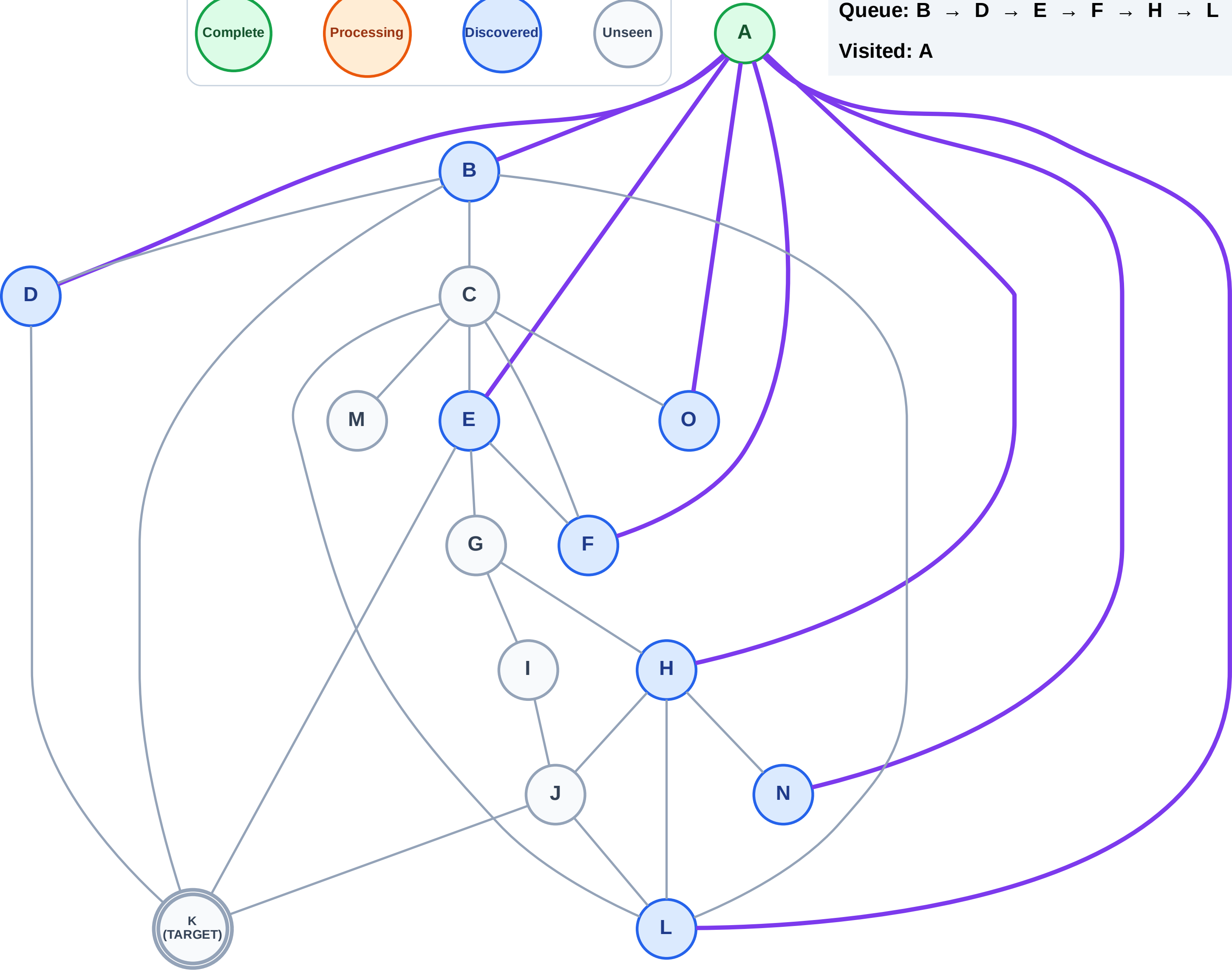
Processing

Discovered

Unseen

Queue: B → D → E → F → H → L → N → O

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

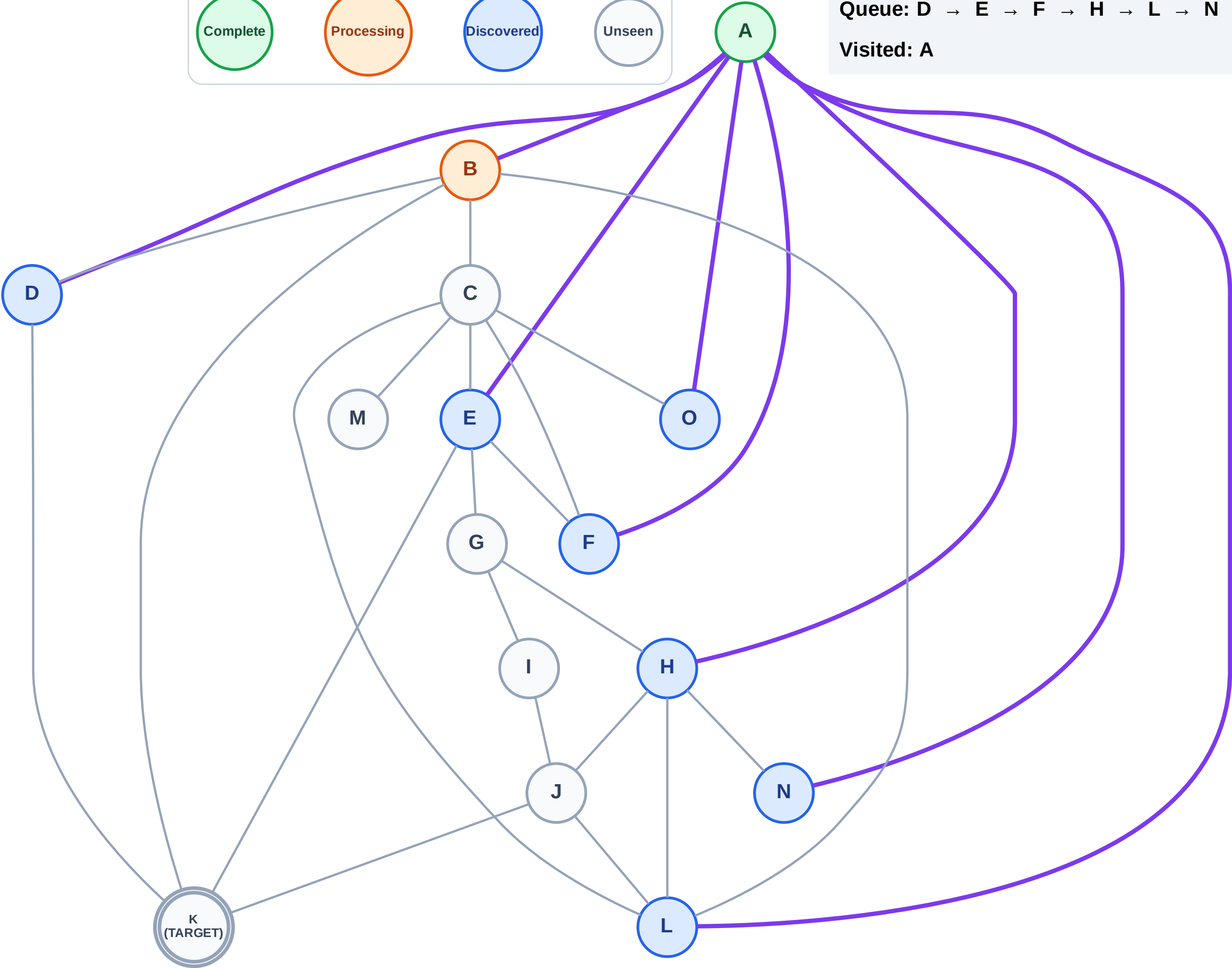
Processing

Discovered

Unseen

Queue: D → E → F → H → L → N → O

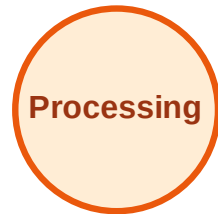
Visited: A



# BFS Traversal

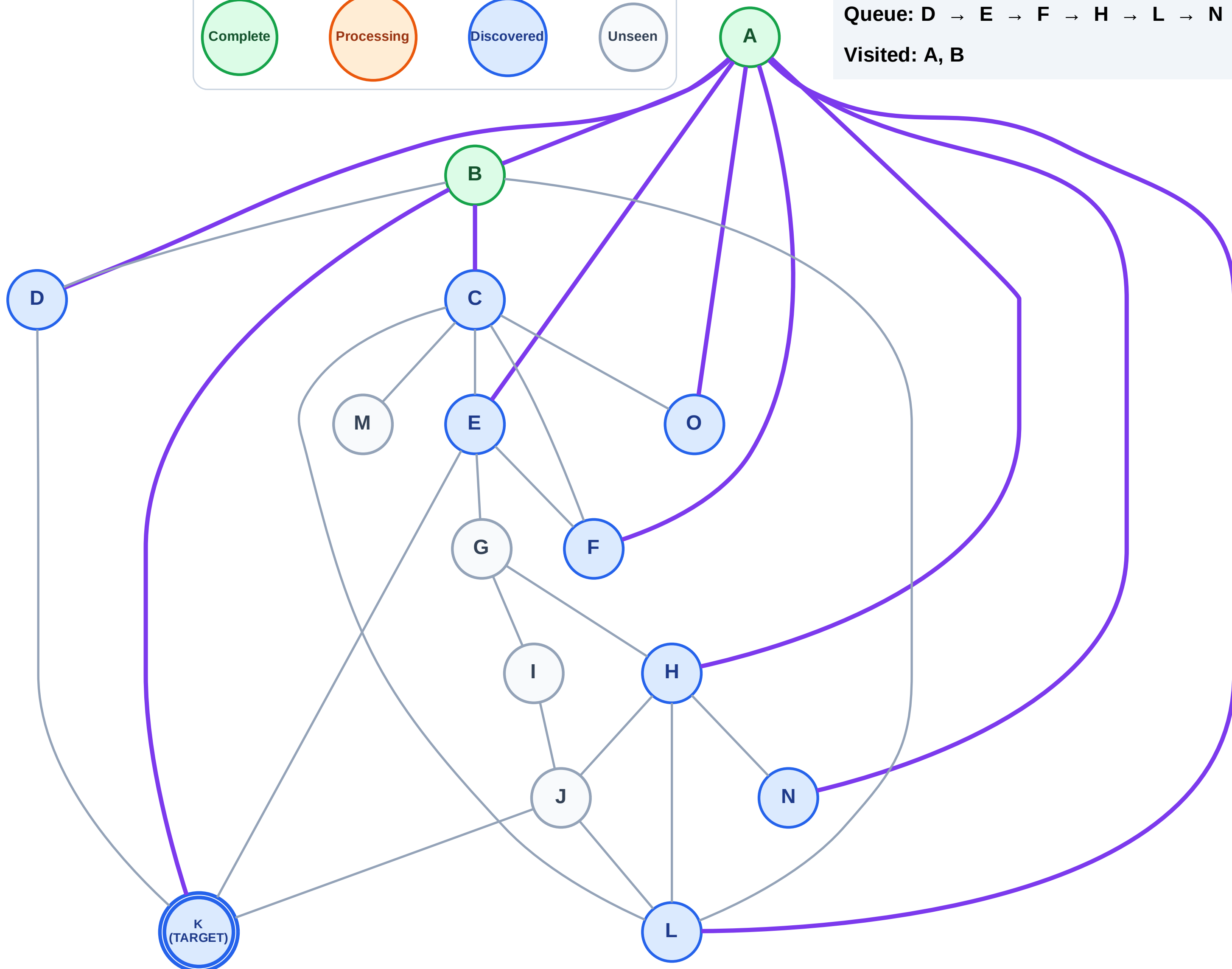
Step 5: Discover C, K from B

## Legend



Queue: D → E → F → H → L → N → O → C → K

Visited: A, B



# BFS Traversal

Step 6: Process node D

Legend

Complete

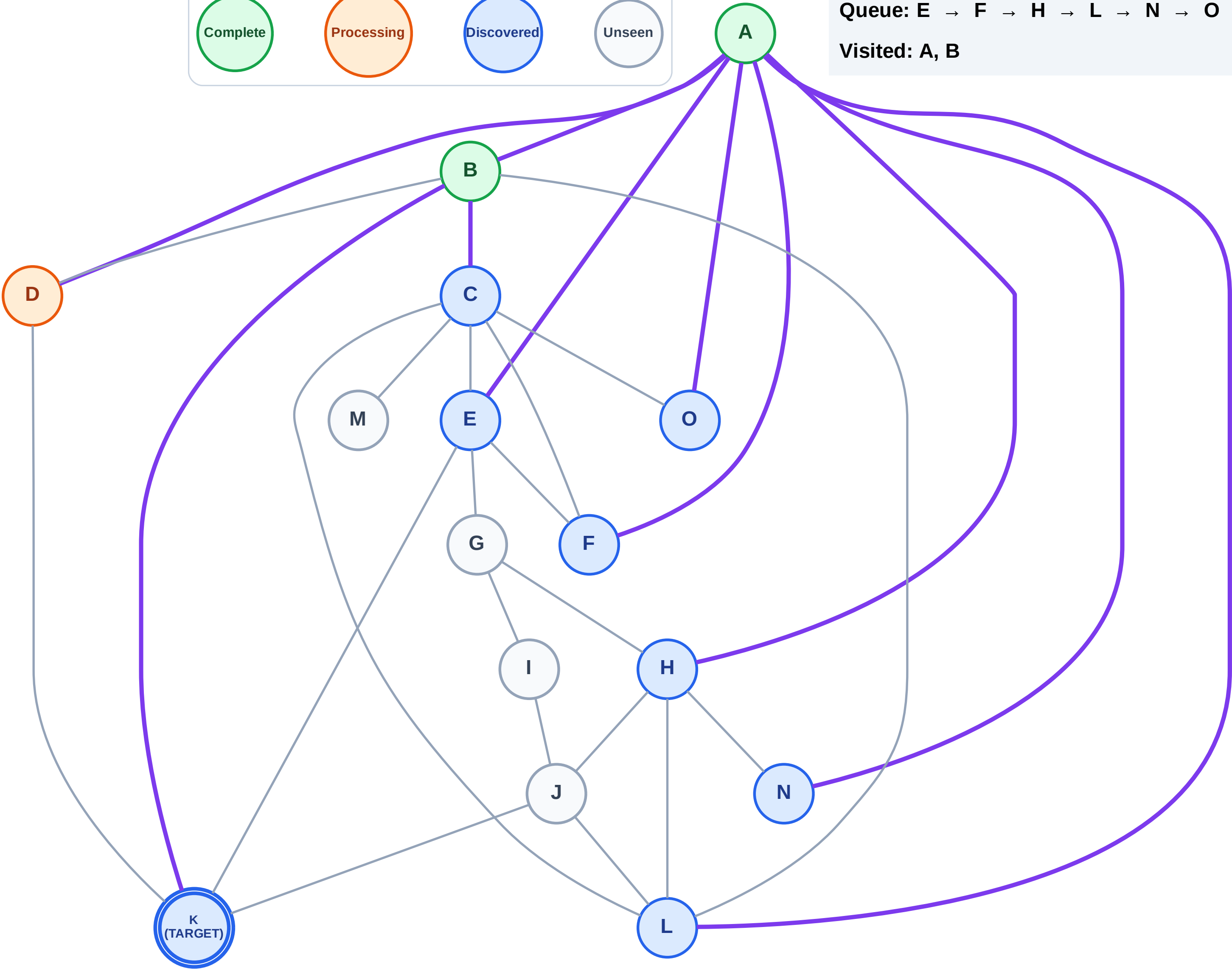
Processing

Discovered

Unseen

Queue: E → F → H → L → N → O → C → K

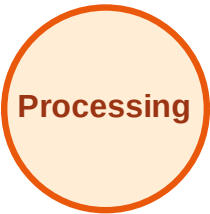
Visited: A, B



# BFS Traversal

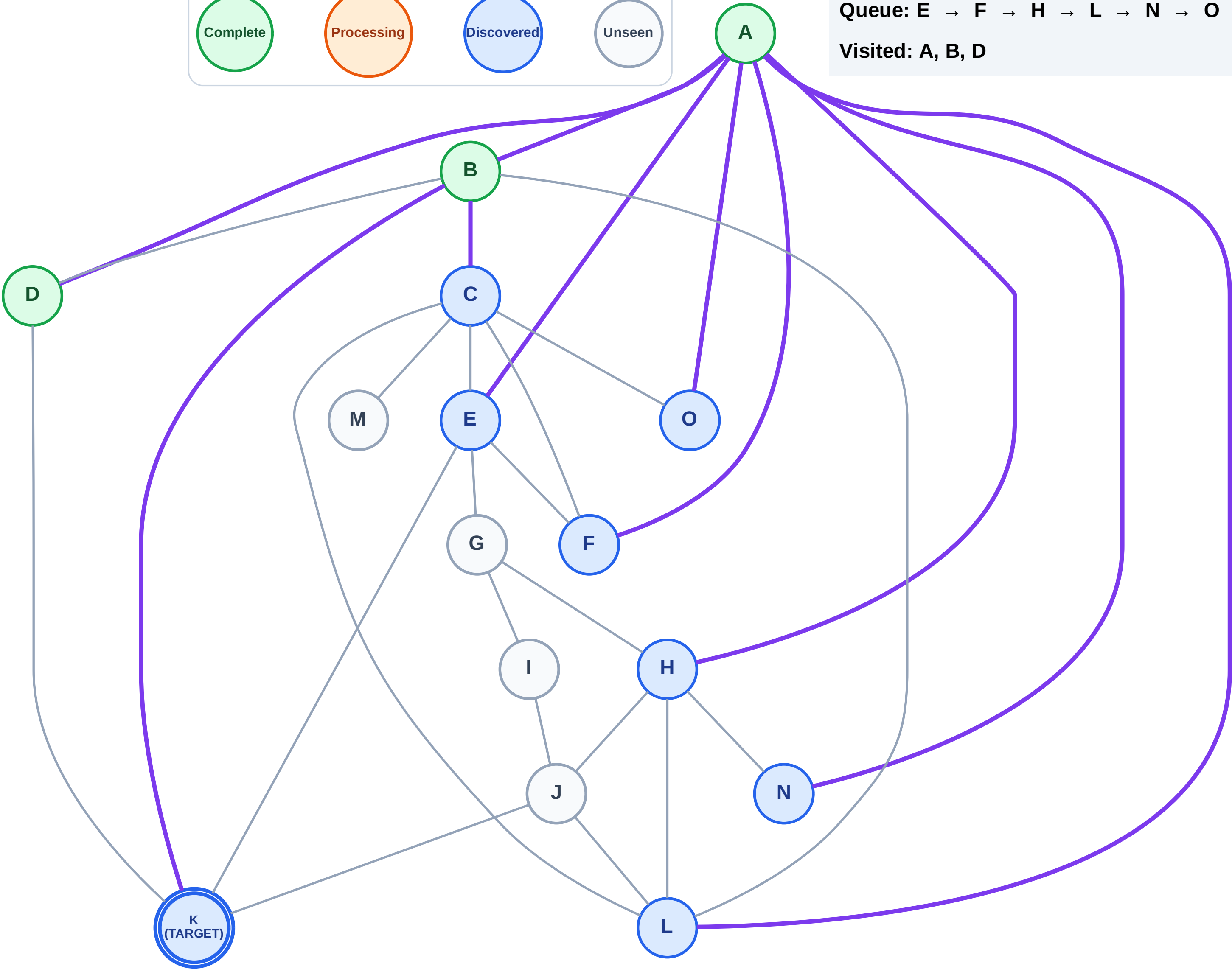
Step 7: No new neighbors from D

## Legend



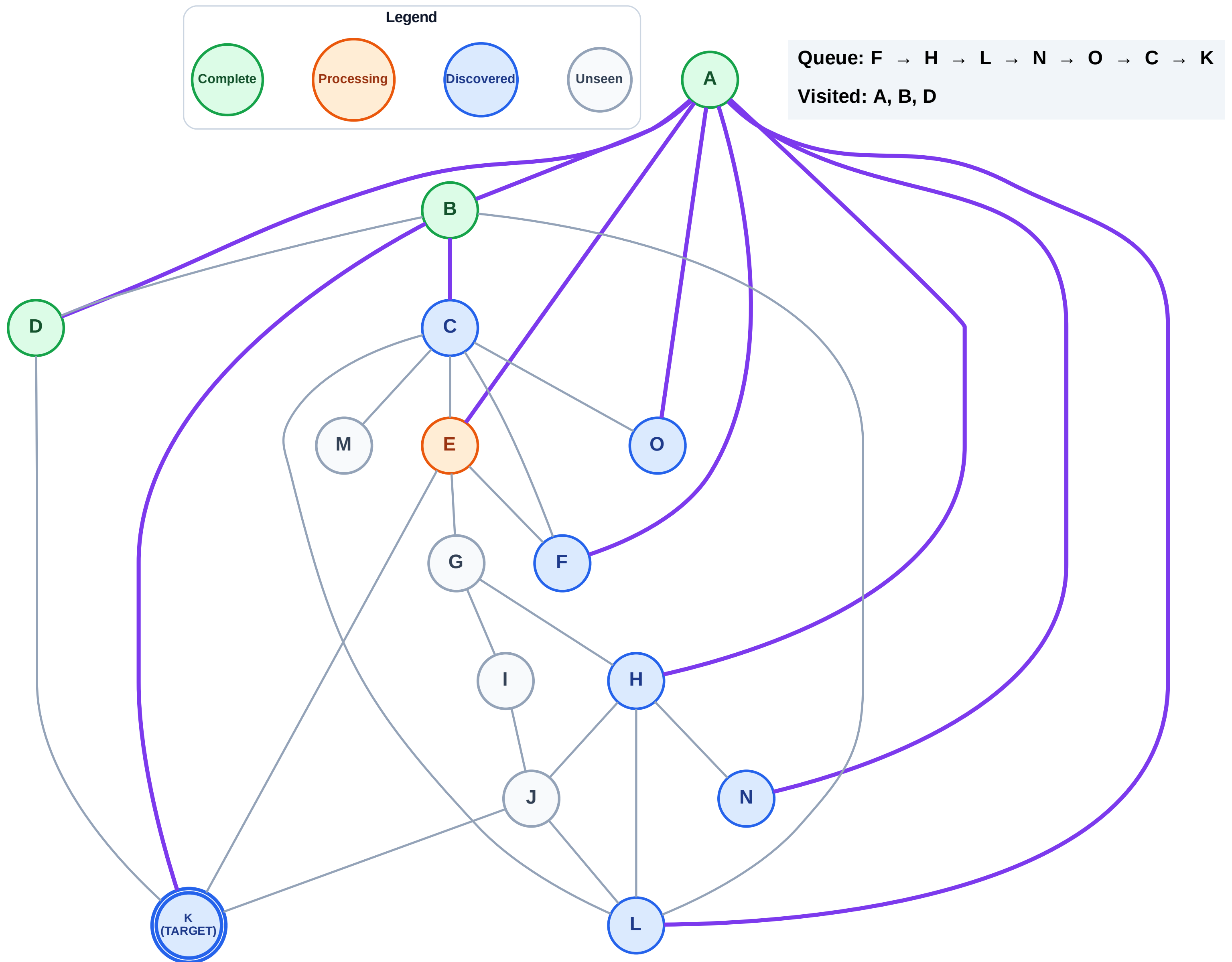
Queue: E → F → H → L → N → O → C → K

Visited: A, B, D



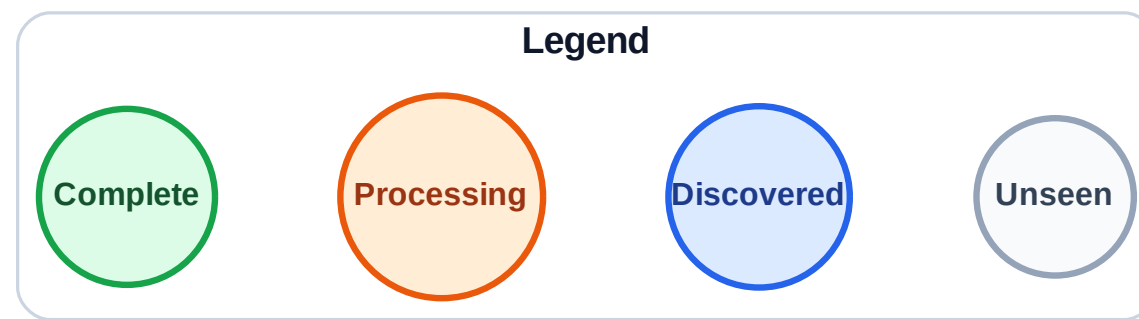
### Step 8: Process node E

### Step 8: Process node E



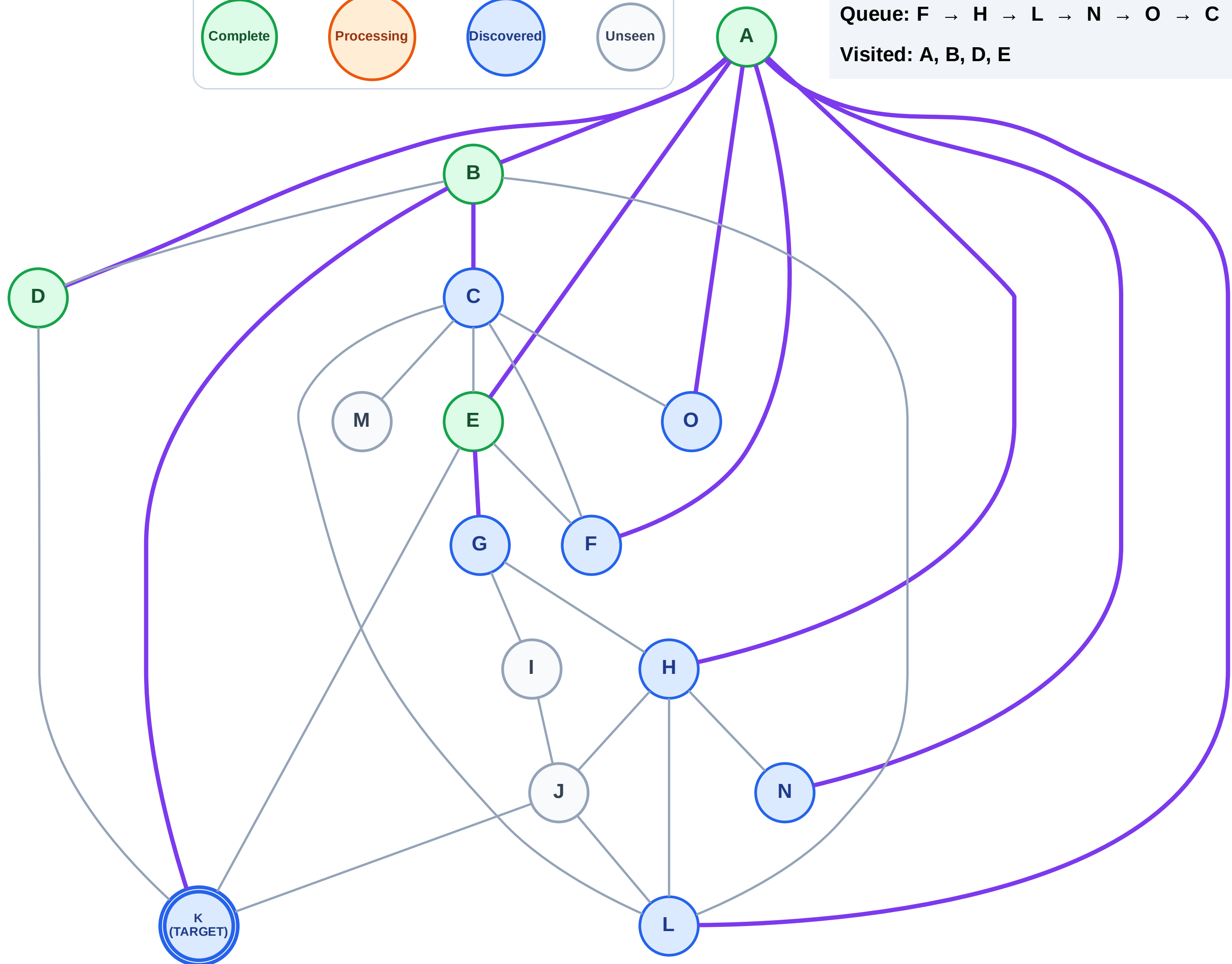
# BFS Traversal

Step 9: Discover G from E



Queue: F → H → L → N → O → C → K → G

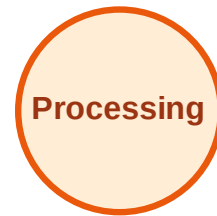
Visited: A, B, D, E



# BFS Traversal

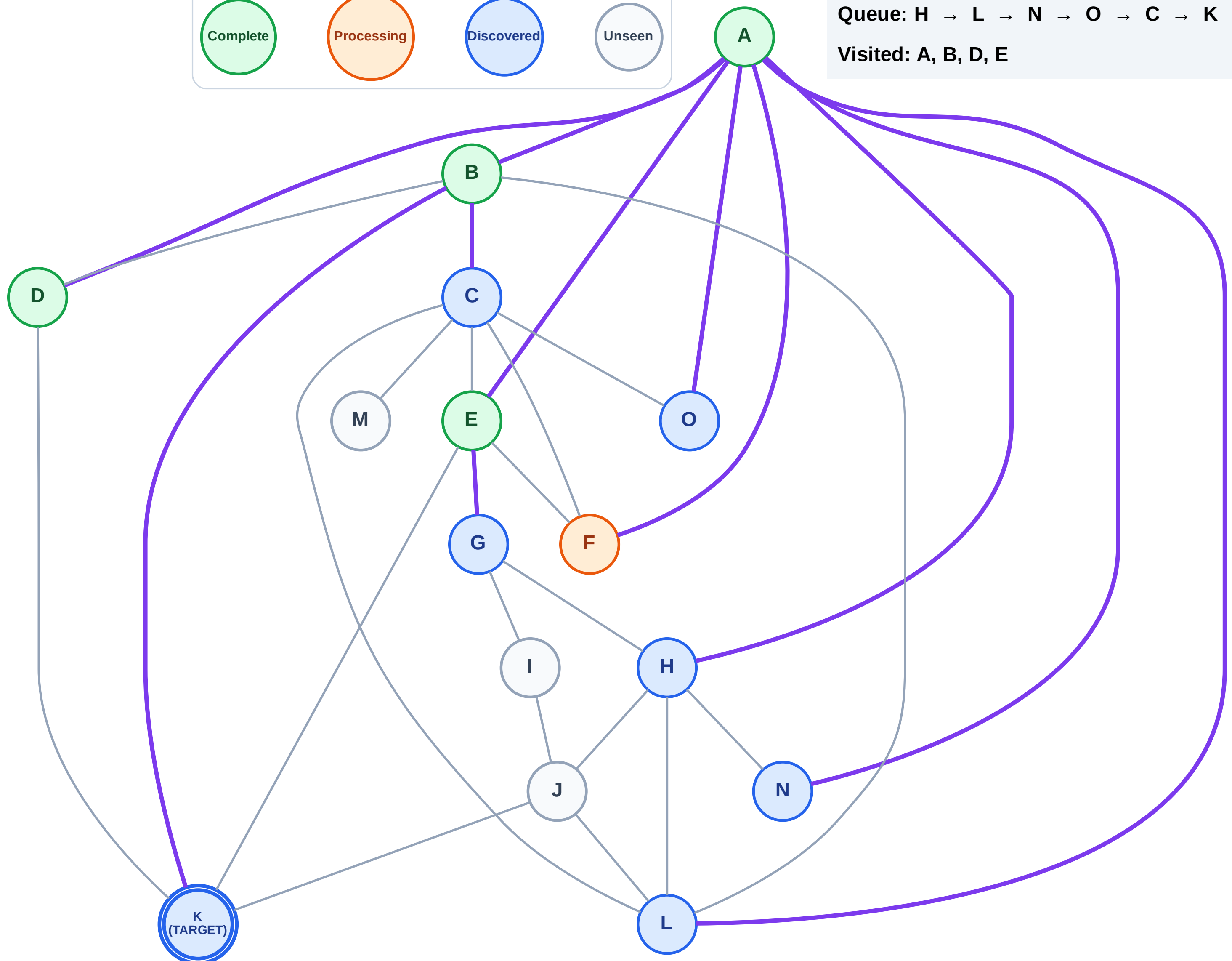
Step 10: Process node F

## Legend



Queue: H → L → N → O → C → K → G

Visited: A, B, D, E





# BFS Traversal

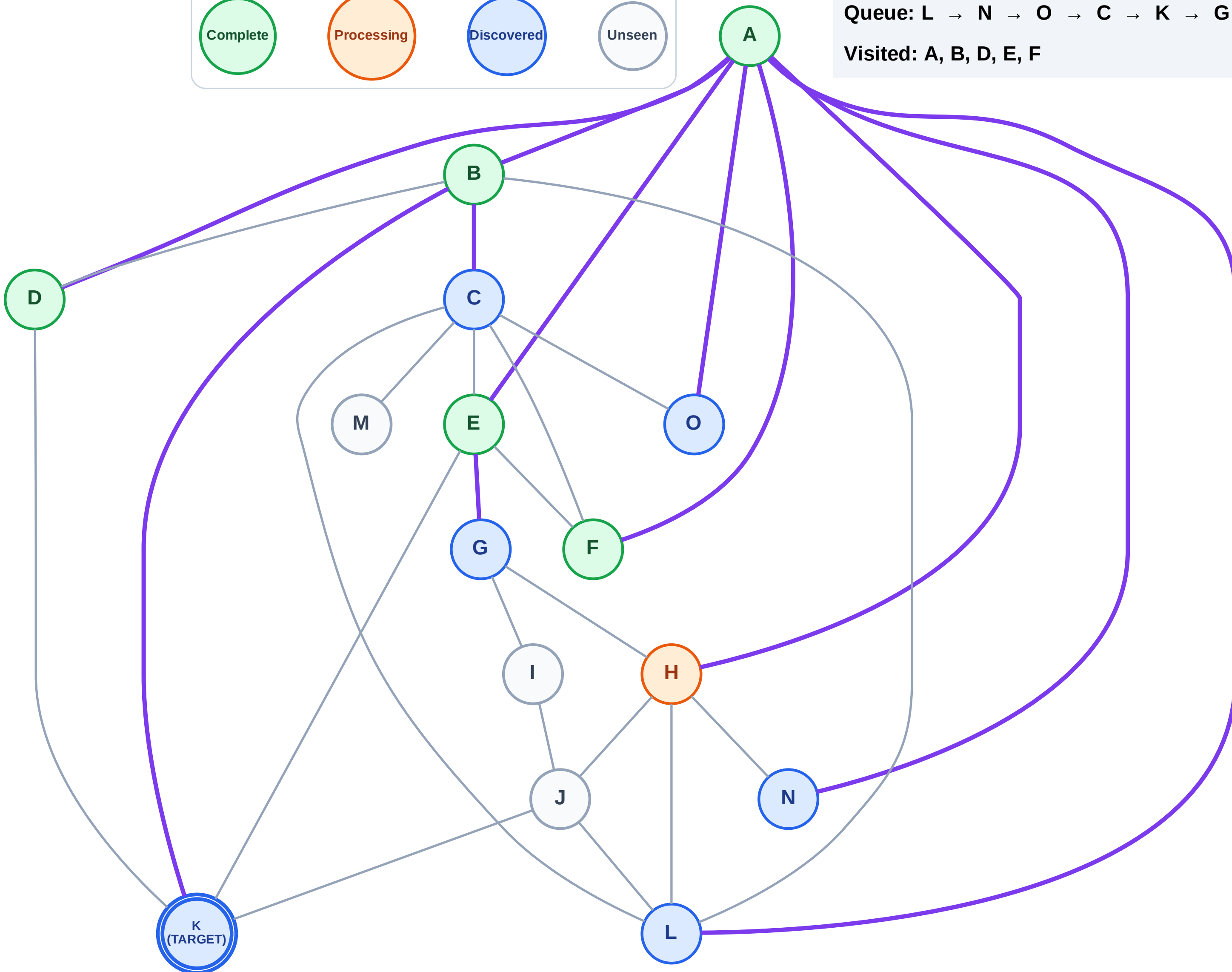
Step 12: Process node H

## Legend



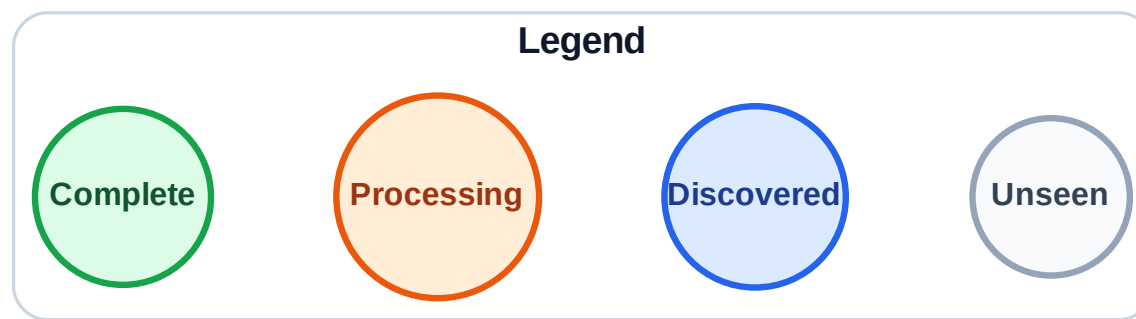
Queue: L → N → O → C → K → G

Visited: A, B, D, E, F



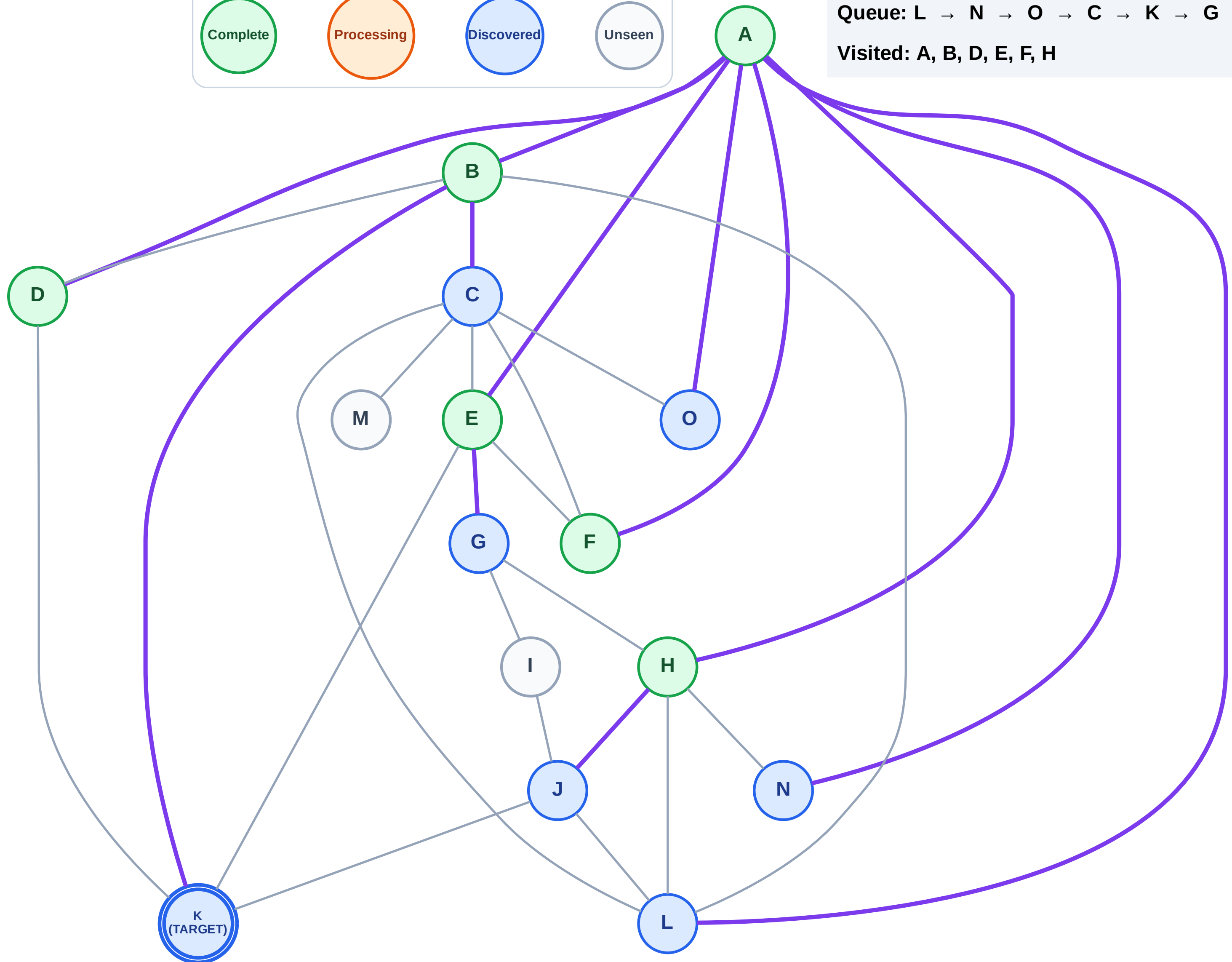
# BFS Traversal

Step 13: Discover J from H



Queue: L → N → O → C → K → G → J

Visited: A, B, D, E, F, H



# BFS Traversal

Step 14: Process node L

Legend

Complete

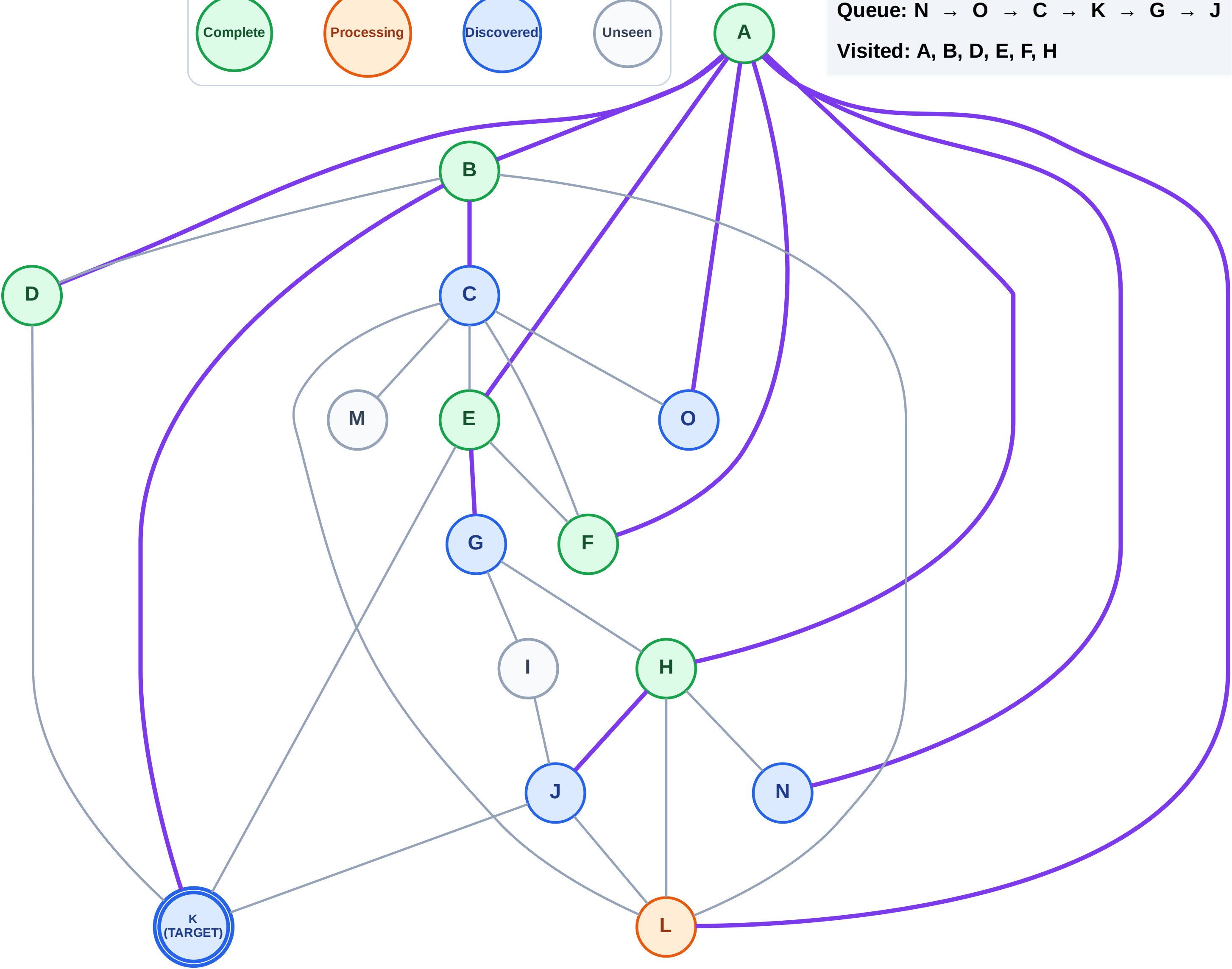
Processing

Discovered

Unseen

Queue: N → O → C → K → G → J

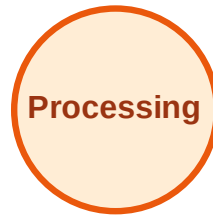
Visited: A, B, D, E, F, H



# BFS Traversal

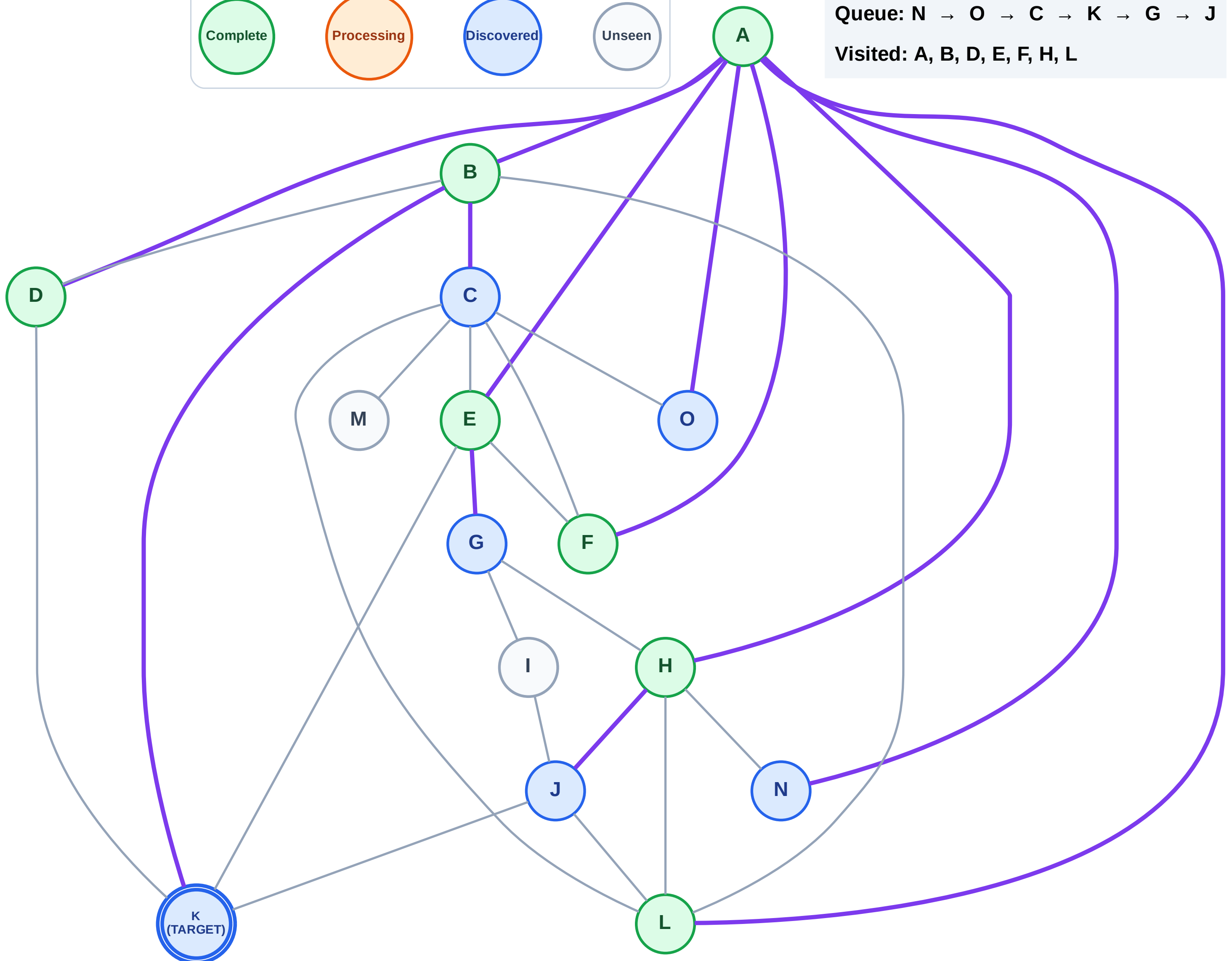
Step 15: No new neighbors from L

## Legend



Queue: N → O → C → K → G → J

Visited: A, B, D, E, F, H, L



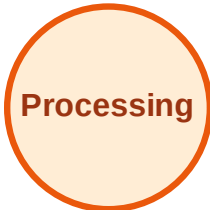
# BFS Traversal

Step 16: Process node N

## Legend



Complete



Processing



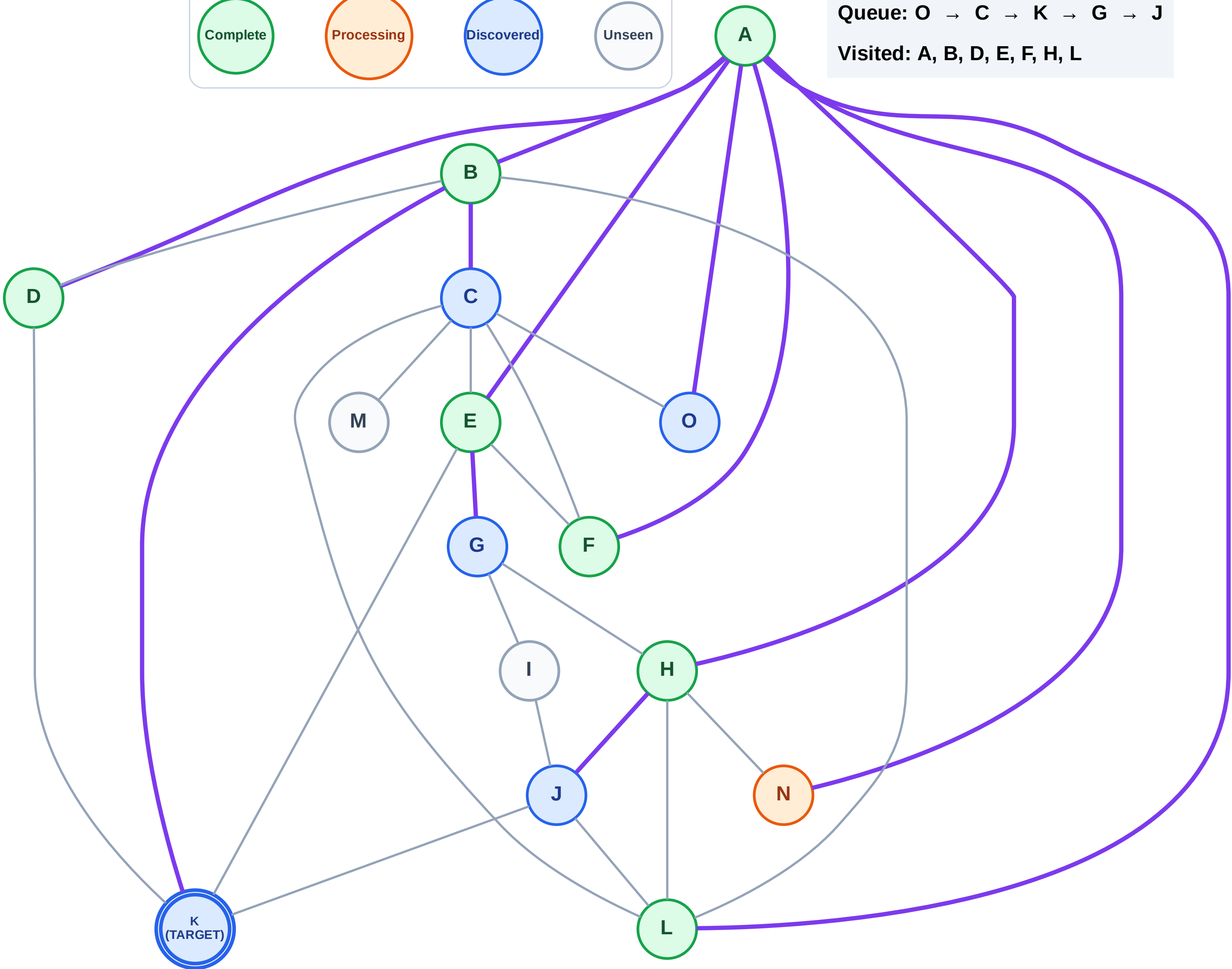
Discovered



Unseen

Queue: O → C → K → G → J

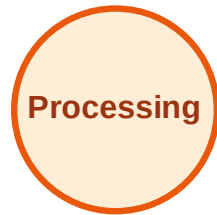
Visited: A, B, D, E, F, H, L



# BFS Traversal

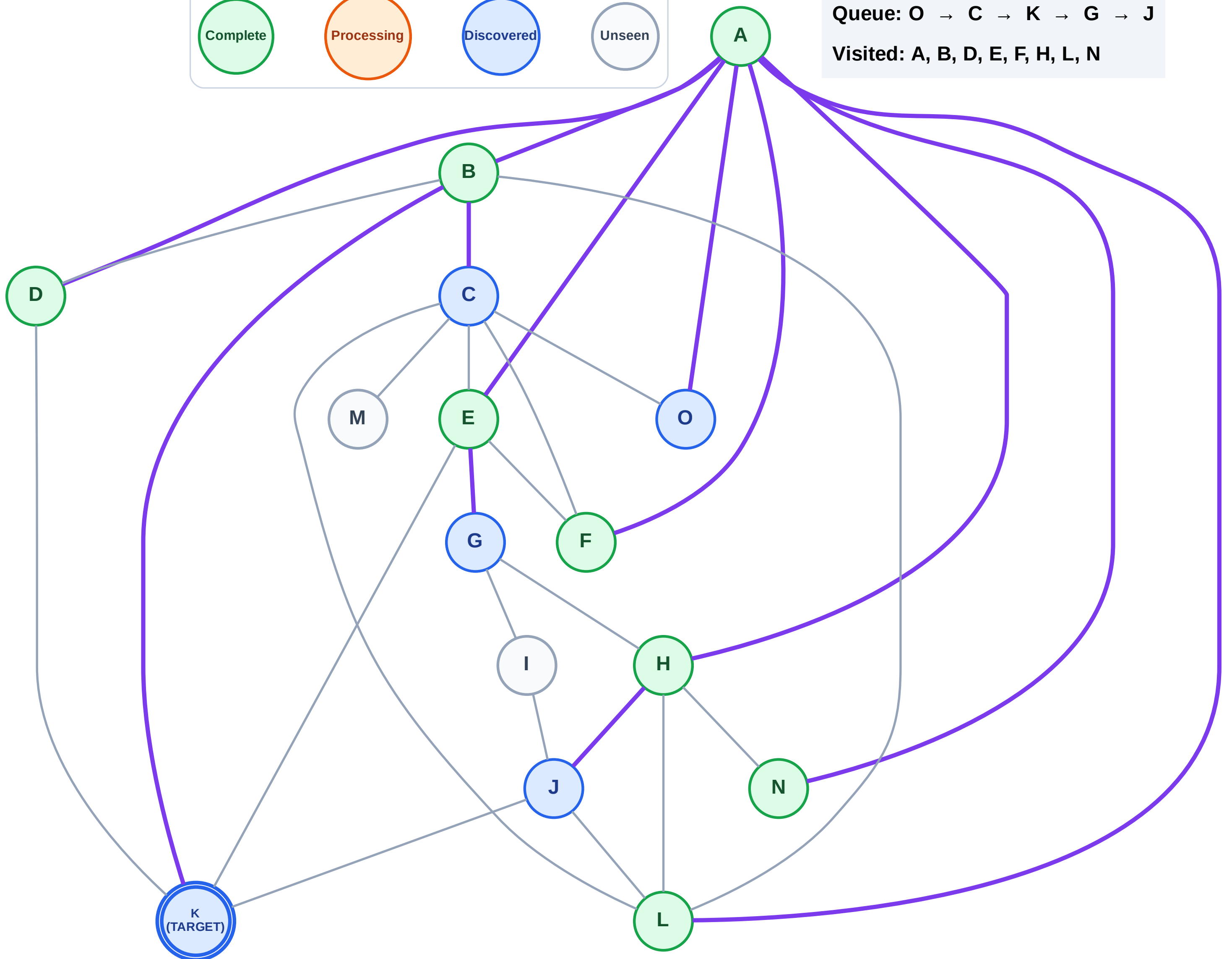
Step 17: No new neighbors from N

## Legend



Queue: O → C → K → G → J

Visited: A, B, D, E, F, H, L, N



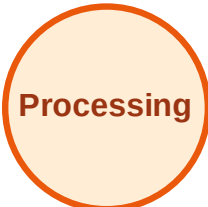
# BFS Traversal

Step 18: Process node O

## Legend



Complete



Processing



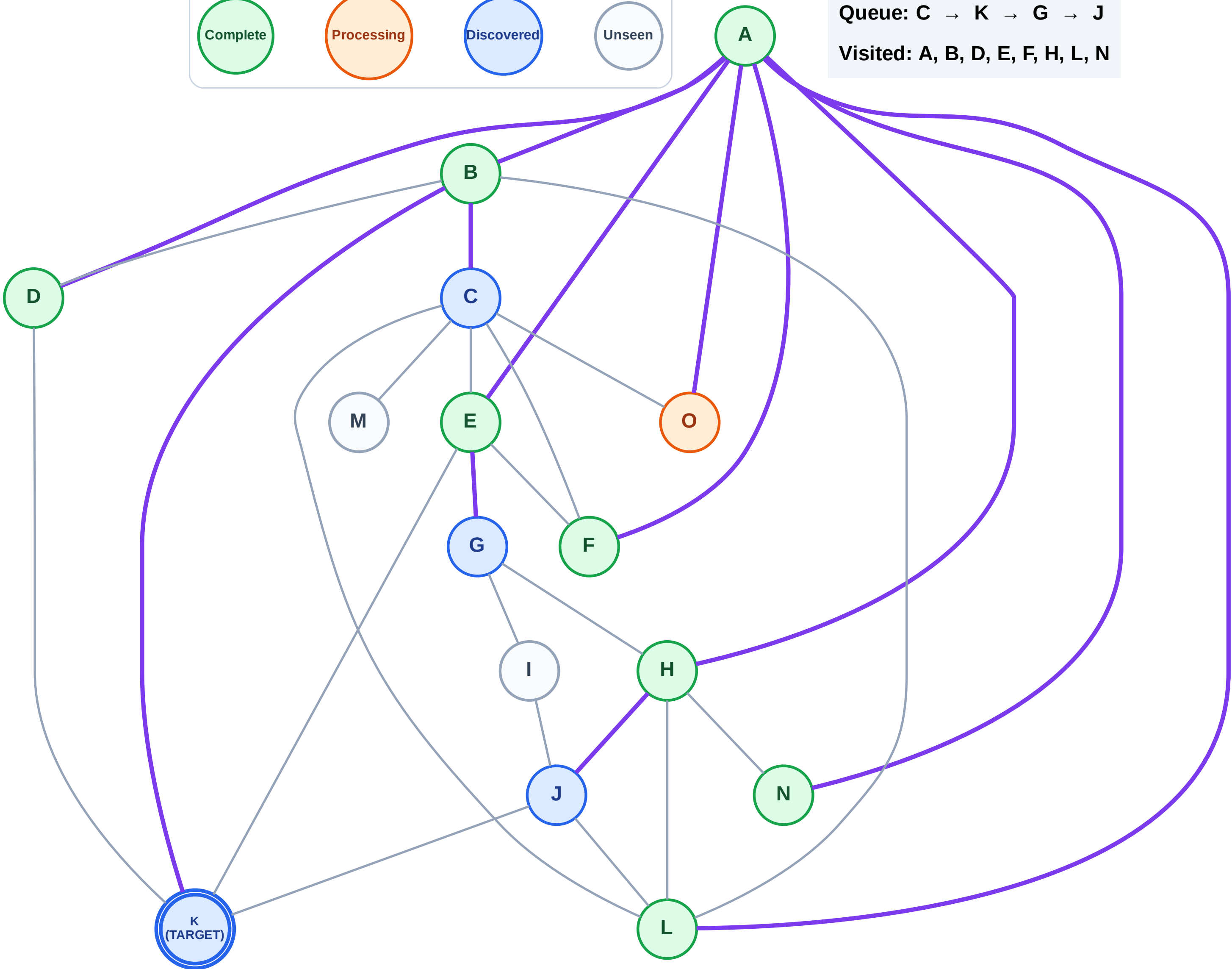
Discovered



Unseen

Queue: C → K → G → J

Visited: A, B, D, E, F, H, L, N



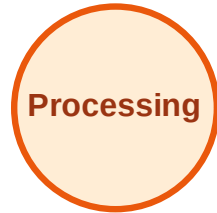
# BFS Traversal

Step 19: No new neighbors from O

## Legend



Complete



Processing



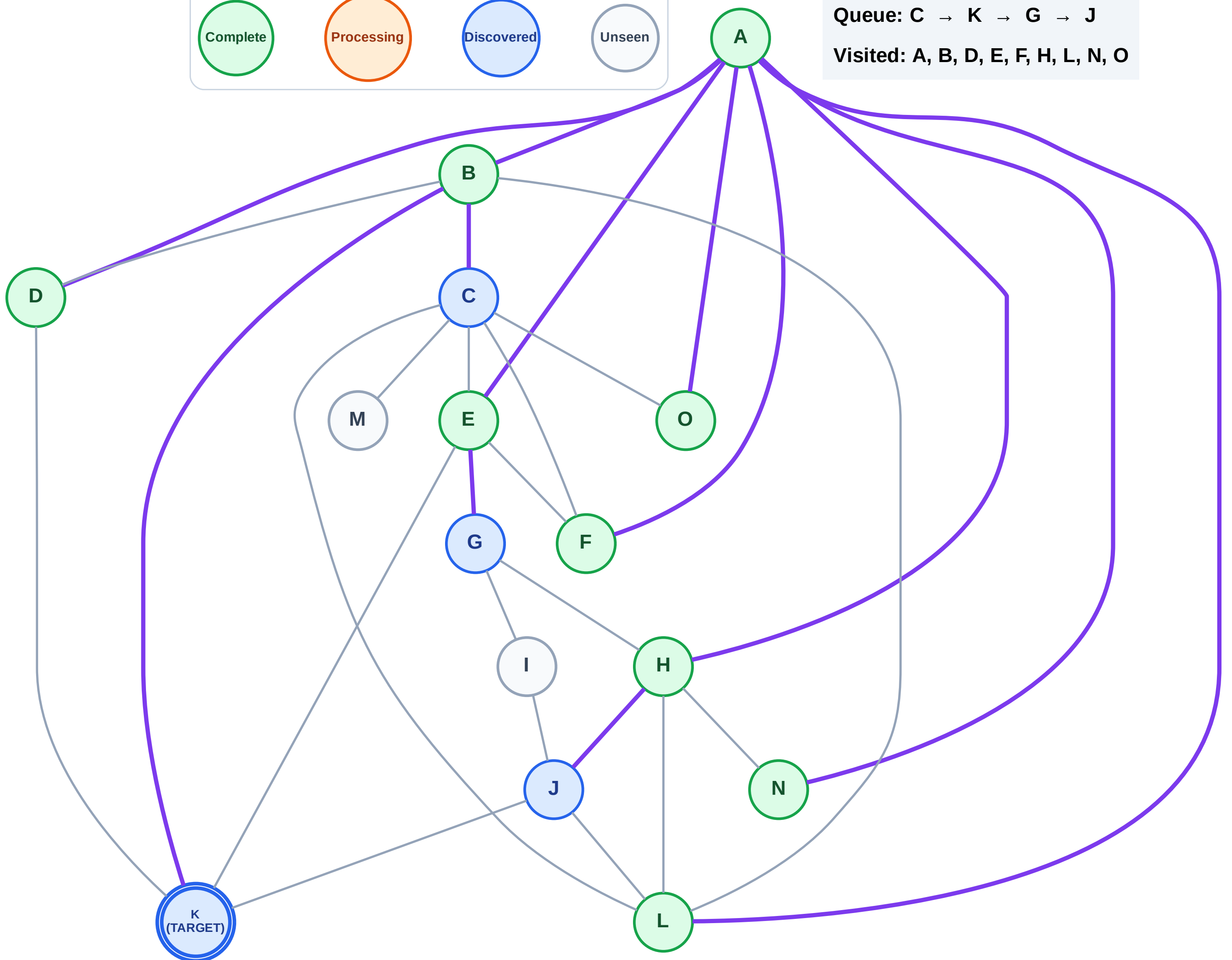
Discovered



Unseen

Queue: C → K → G → J

Visited: A, B, D, E, F, H, L, N, O



# BFS Traversal

Step 20: Process node C

## Legend



Complete



Processing



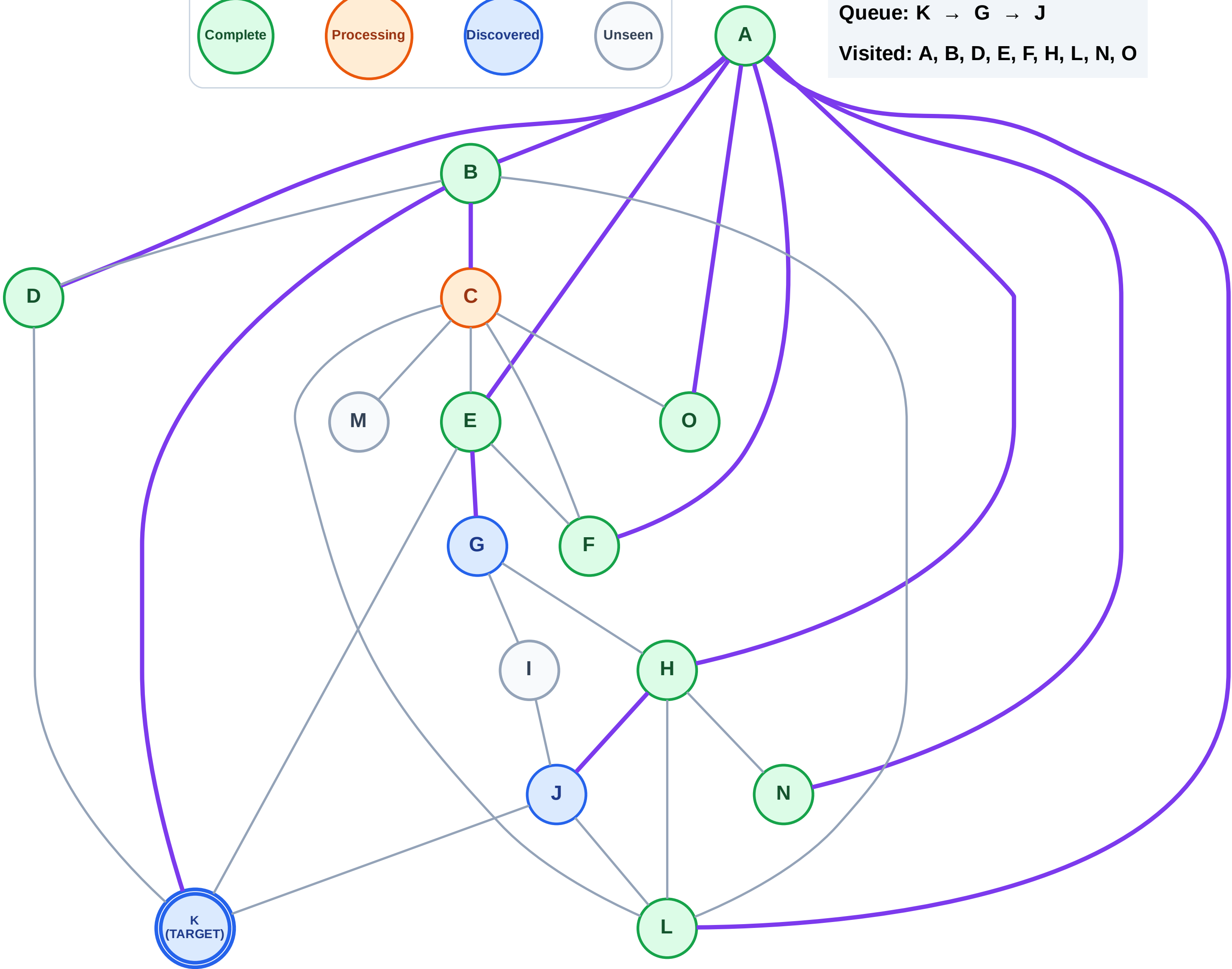
Discovered



Unseen

Queue: K → G → J

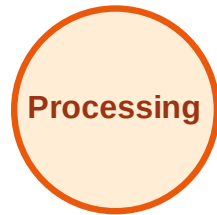
Visited: A, B, D, E, F, H, L, N, O



# BFS Traversal

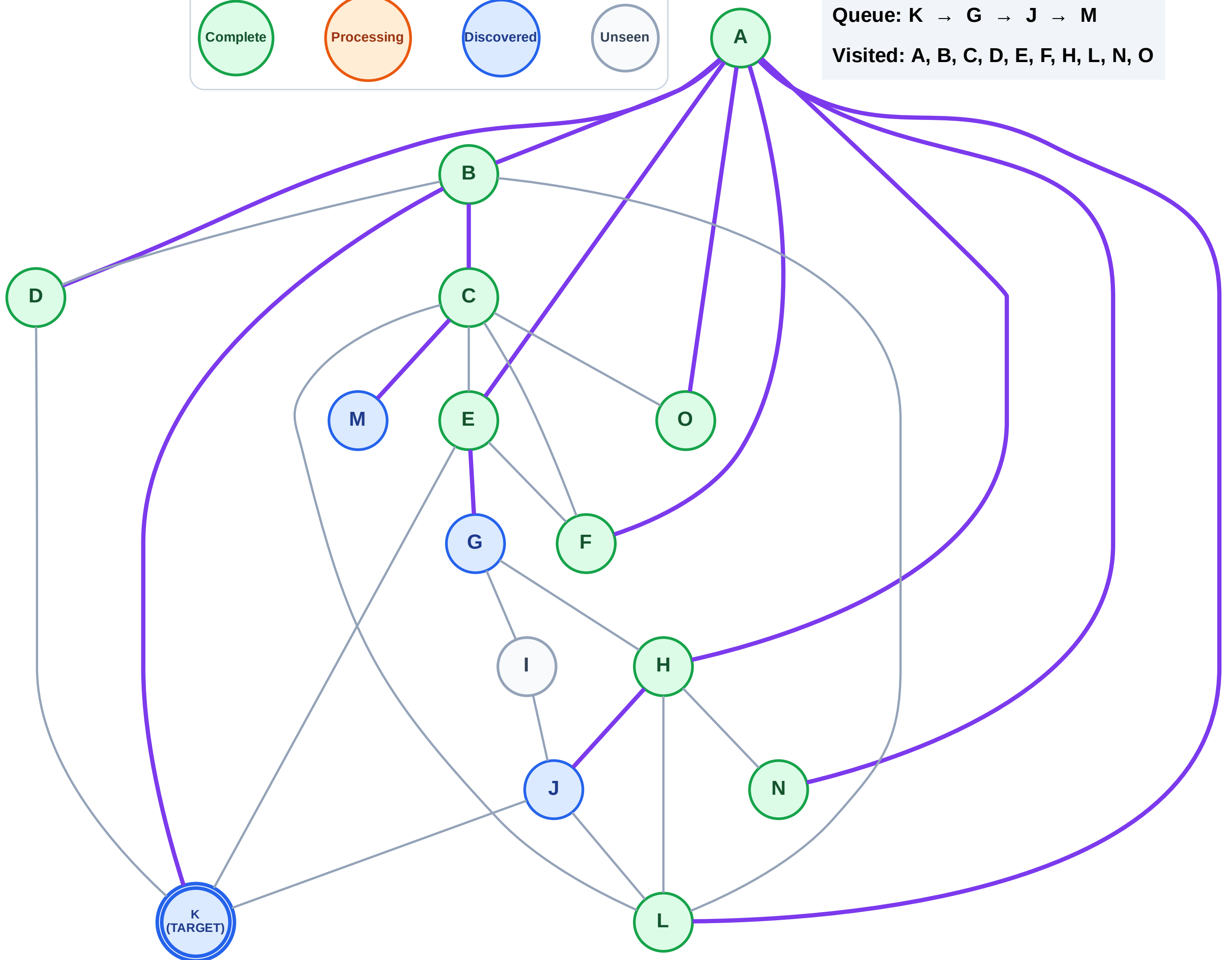
Step 21: Discover M from C

## Legend



Queue: K → G → J → M

Visited: A, B, C, D, E, F, H, L, N, O



# BFS Traversal

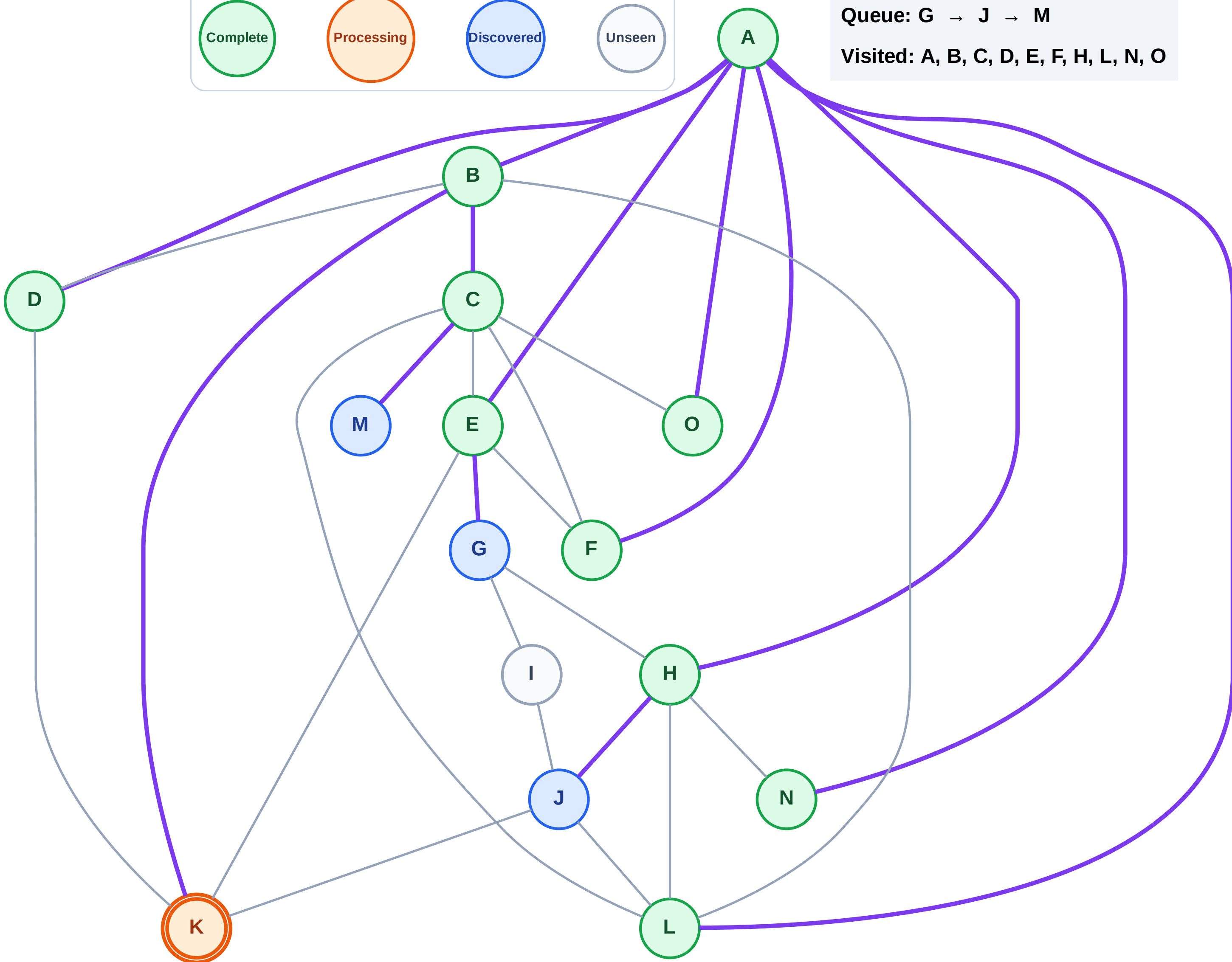
Step 22: Process node K

## Legend



Queue: G → J → M

Visited: A, B, C, D, E, F, H, L, N, O



# BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

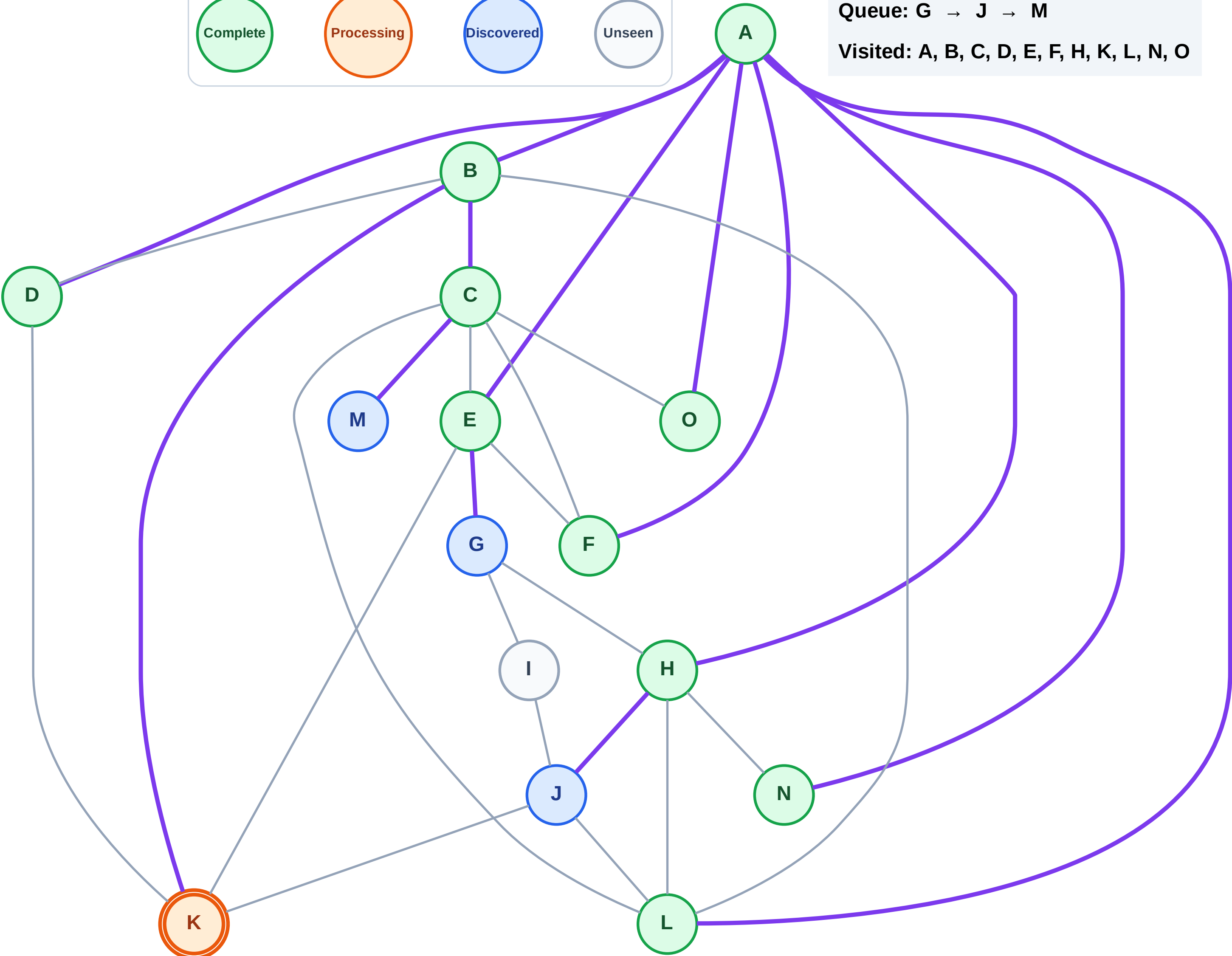
Processing

Discovered

Unseen

Queue: G → J → M

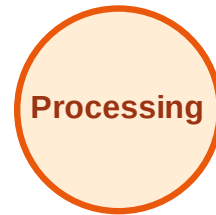
Visited: A, B, C, D, E, F, H, K, L, N, O



# DFS Traversal

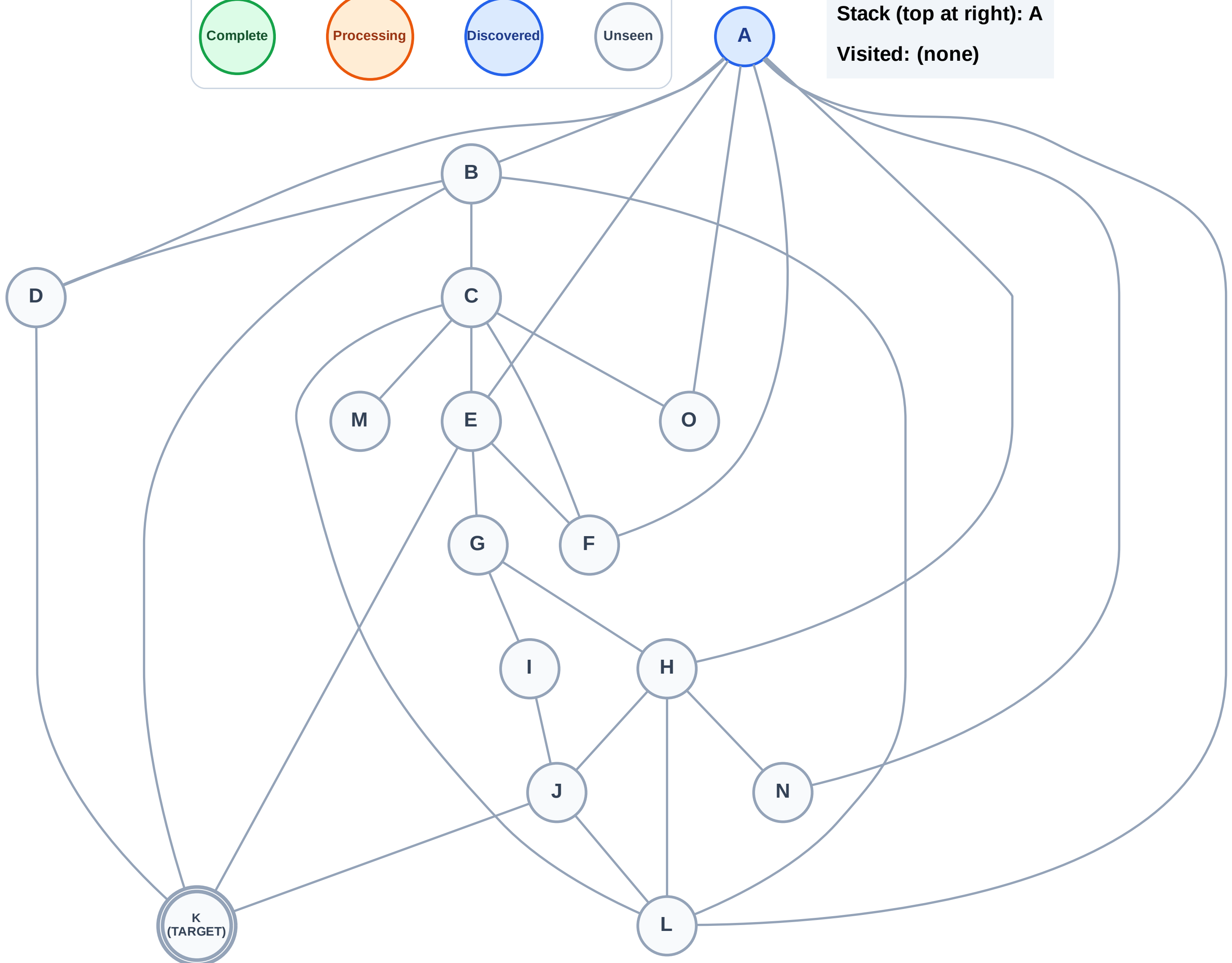
Step 1: Start at A (Seeking target node K)

## Legend



Stack (top at right): A

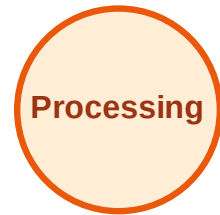
Visited: (none)



# DFS Traversal

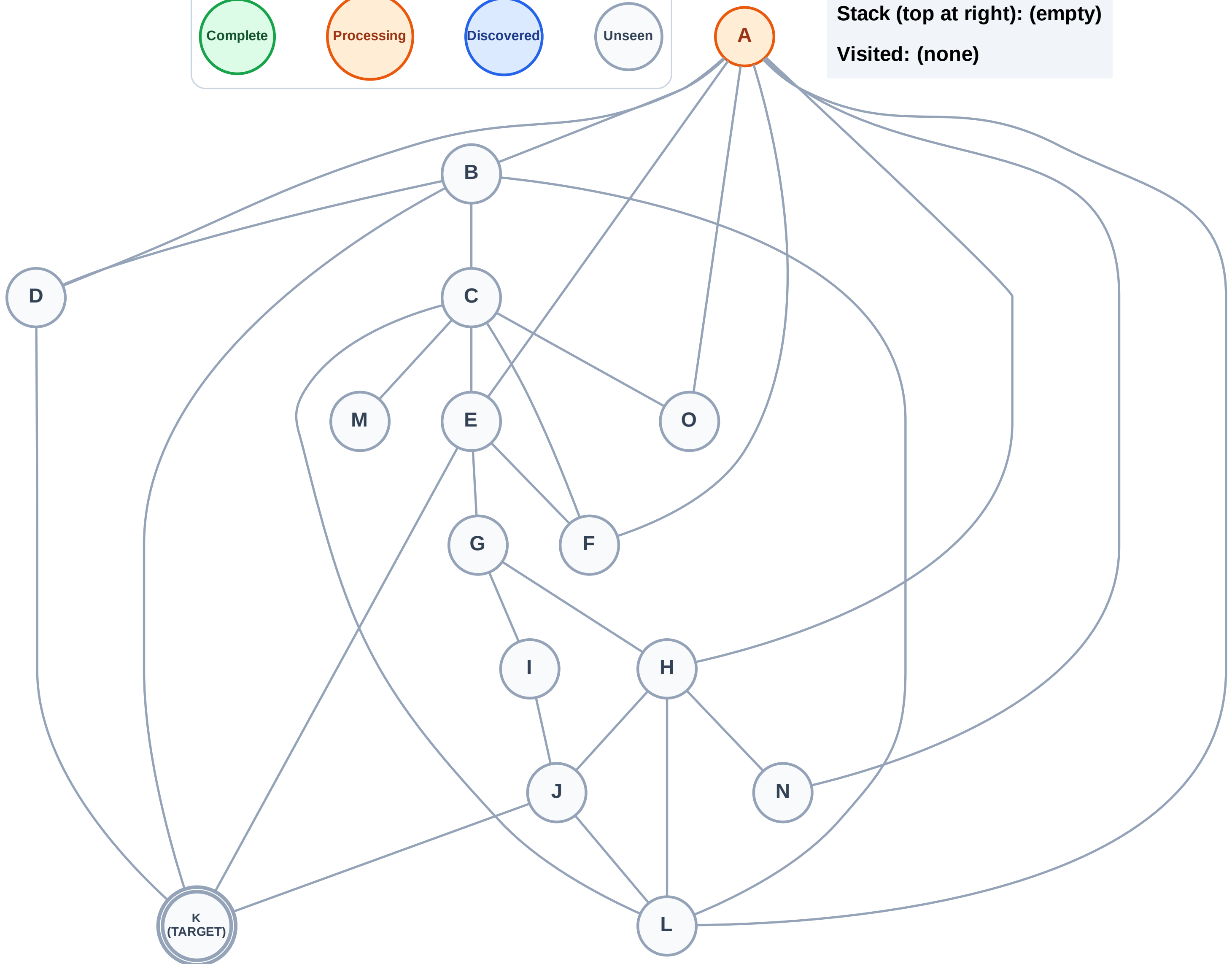
Step 2: Process node A

## Legend



Stack (top at right): (empty)

Visited: (none)



# DFS Traversal

Step 3: Discover O, N, L, H, F, E, D, B from A

Legend

Complete

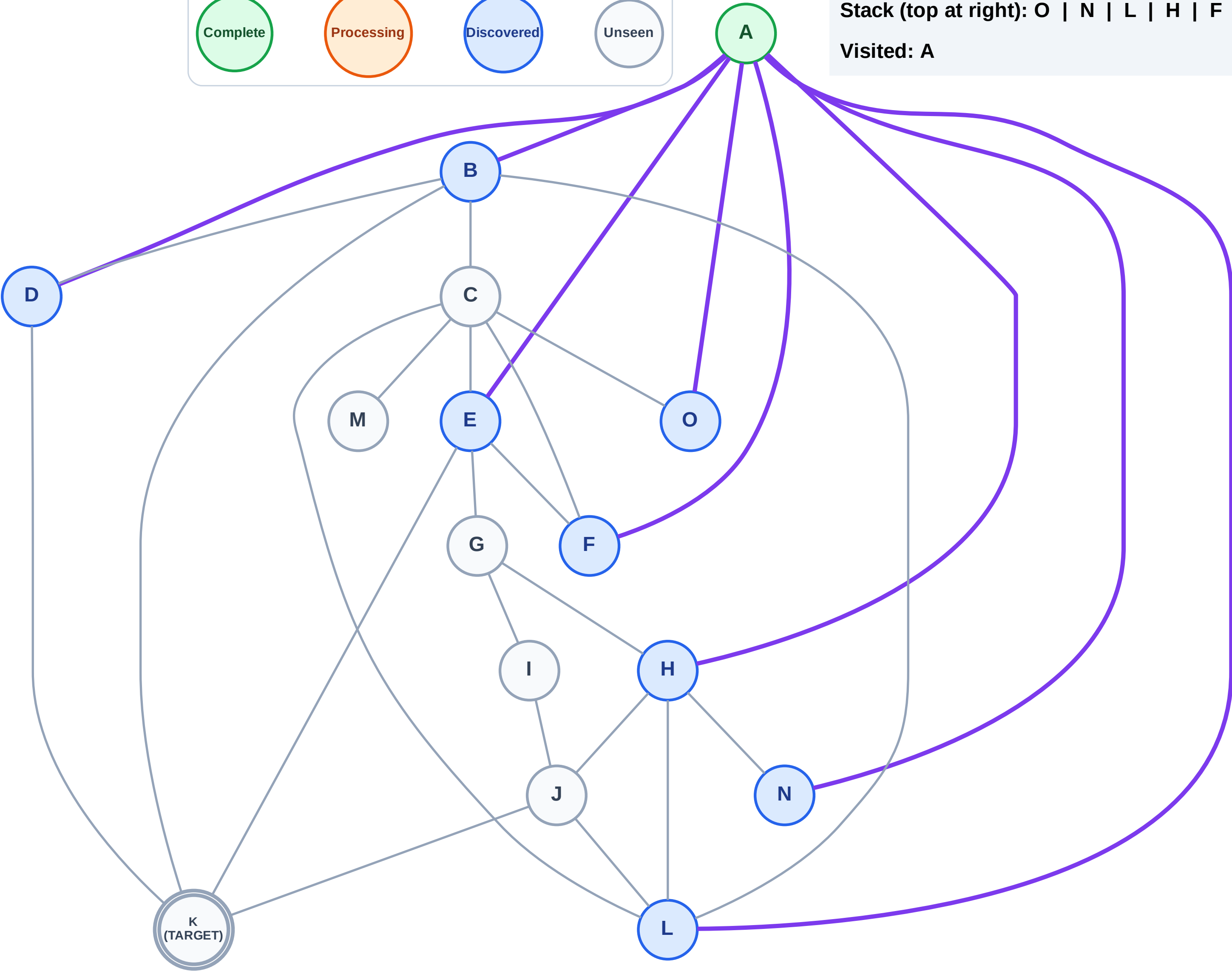
Processing

Discovered

Unseen

Stack (top at right): O | N | L | H | F | E | D | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

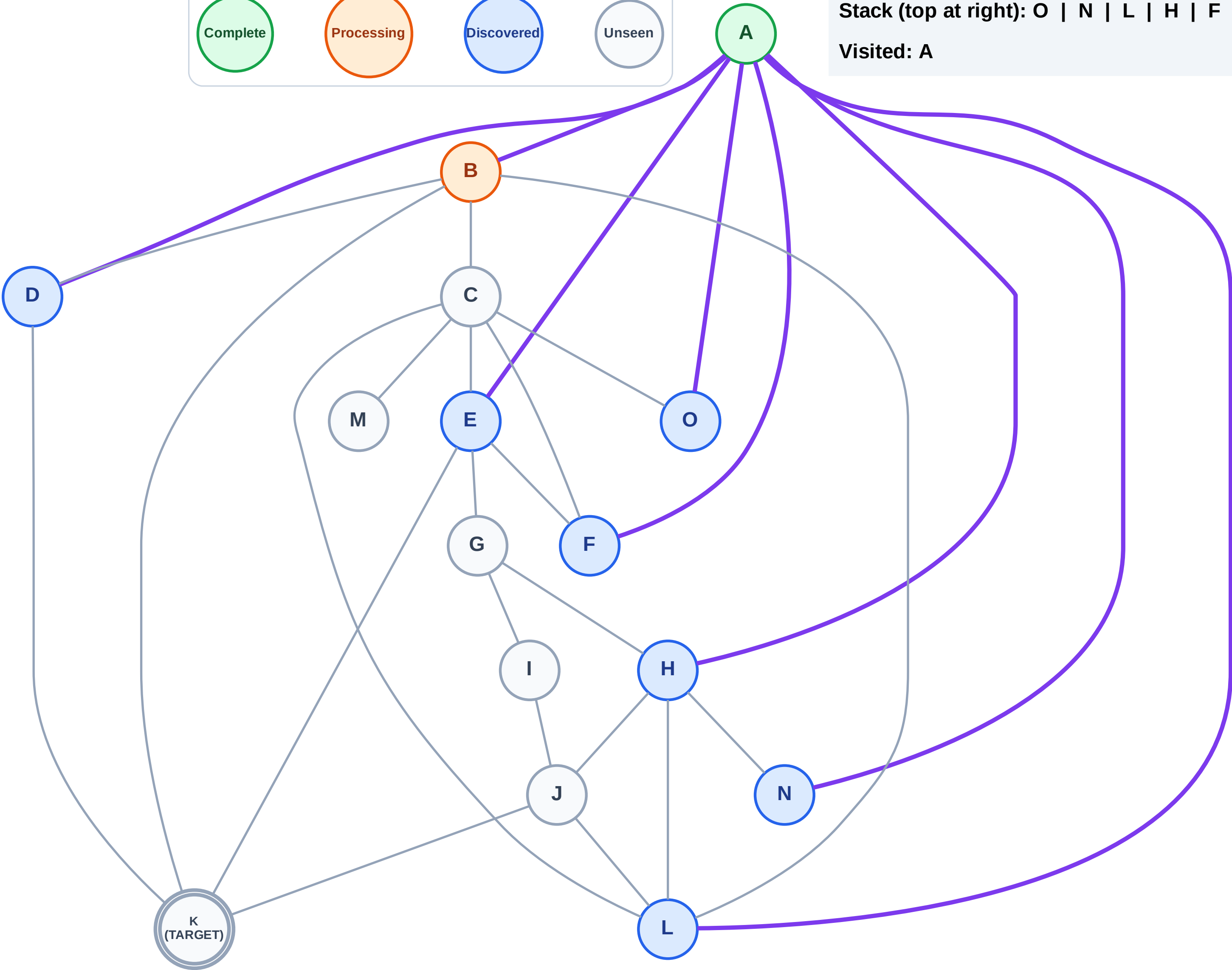
Processing

Discovered

Unseen

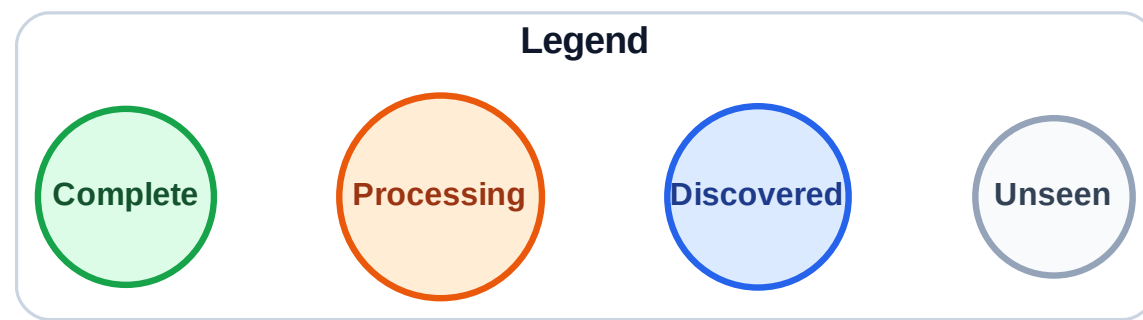
Stack (top at right): O | N | L | H | F | E | D

Visited: A



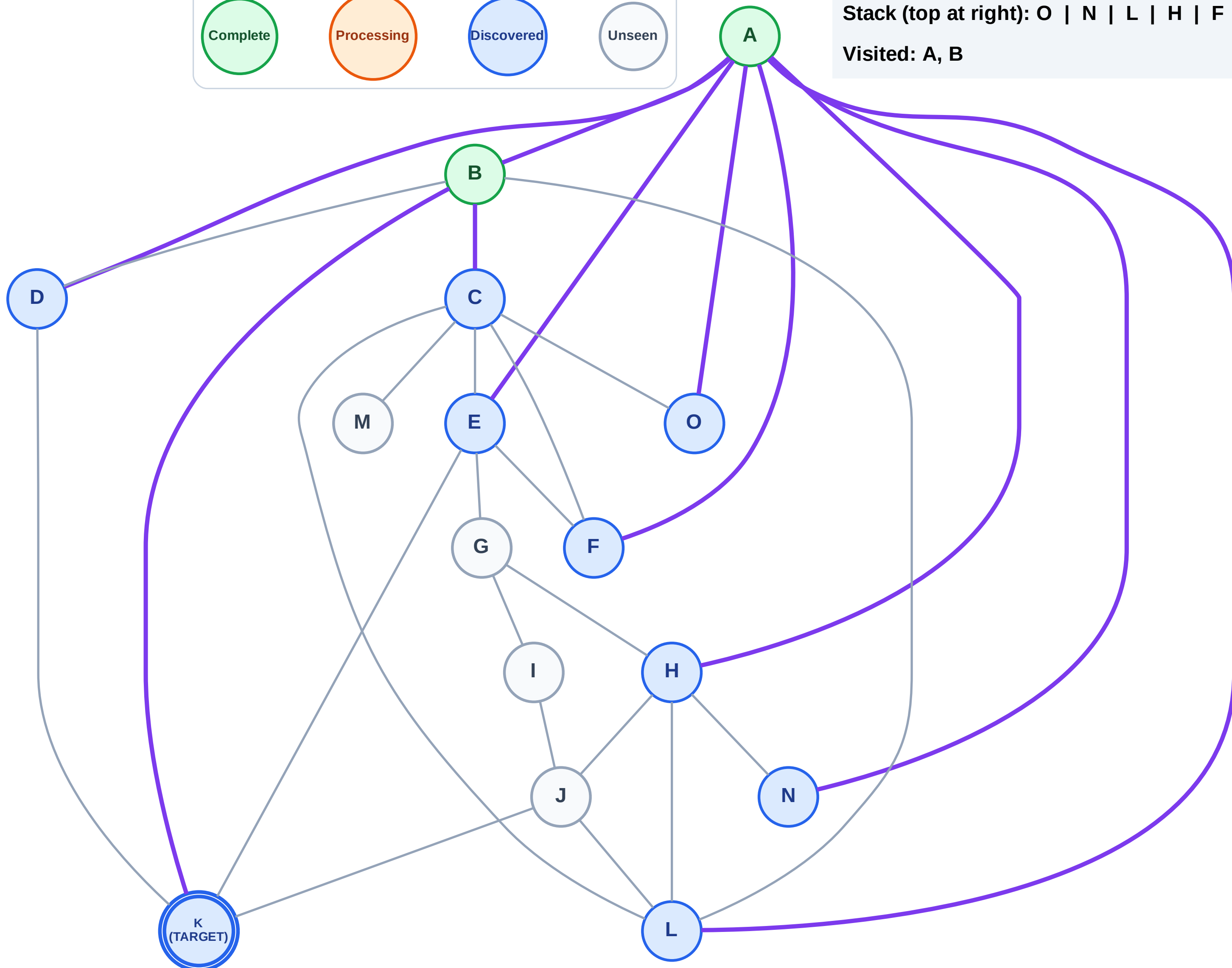
# DFS Traversal

Step 5: Discover K, C from B



Stack (top at right): O | N | L | H | F | E | D | K | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

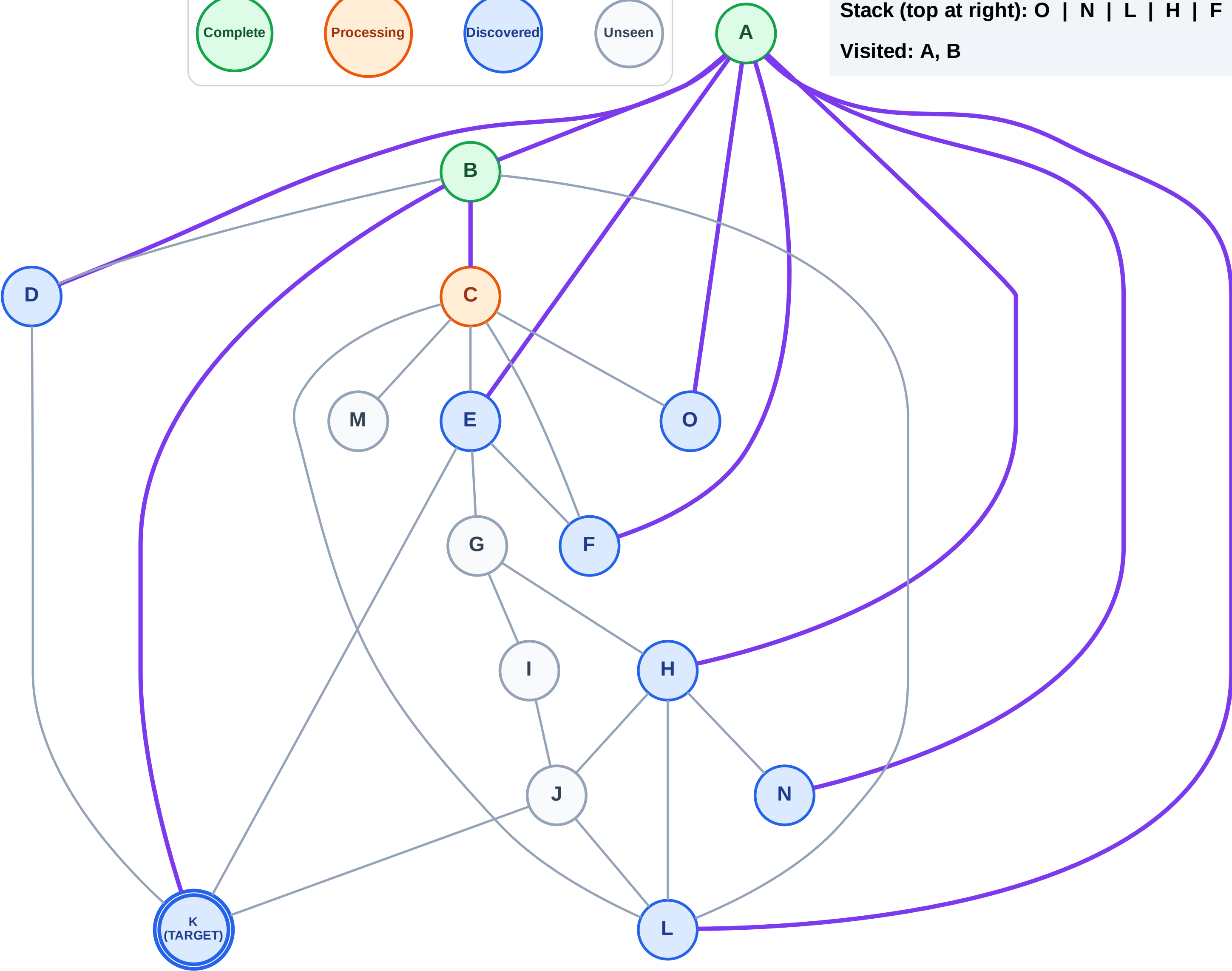
Processing

Discovered

Unseen

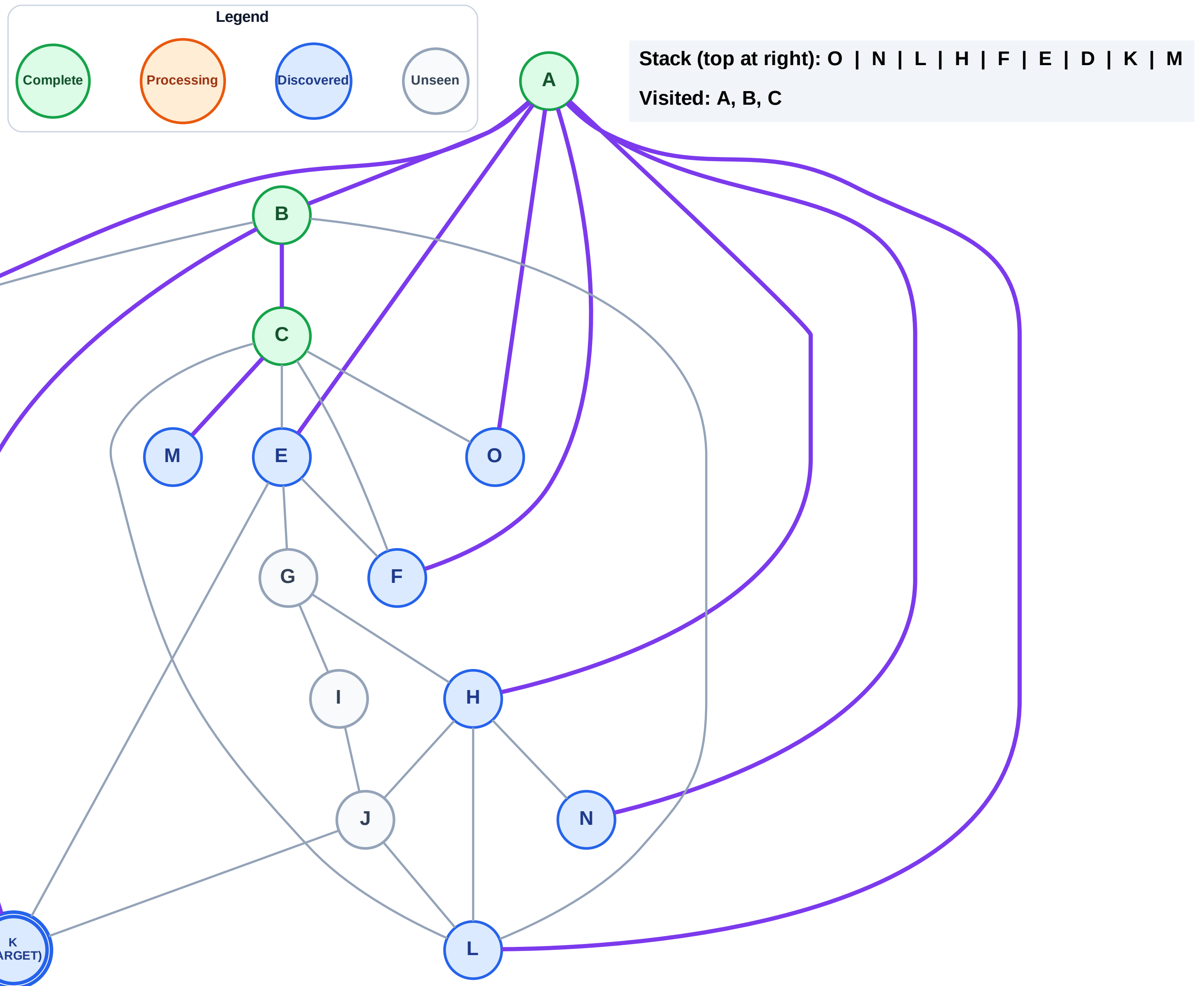
Stack (top at right): O | N | L | H | F | E | D | K

Visited: A, B



### Step 7: Discover M from C

### Step 7: Discover M from C



# DFS Traversal

Step 8: Process node M

Legend

Complete

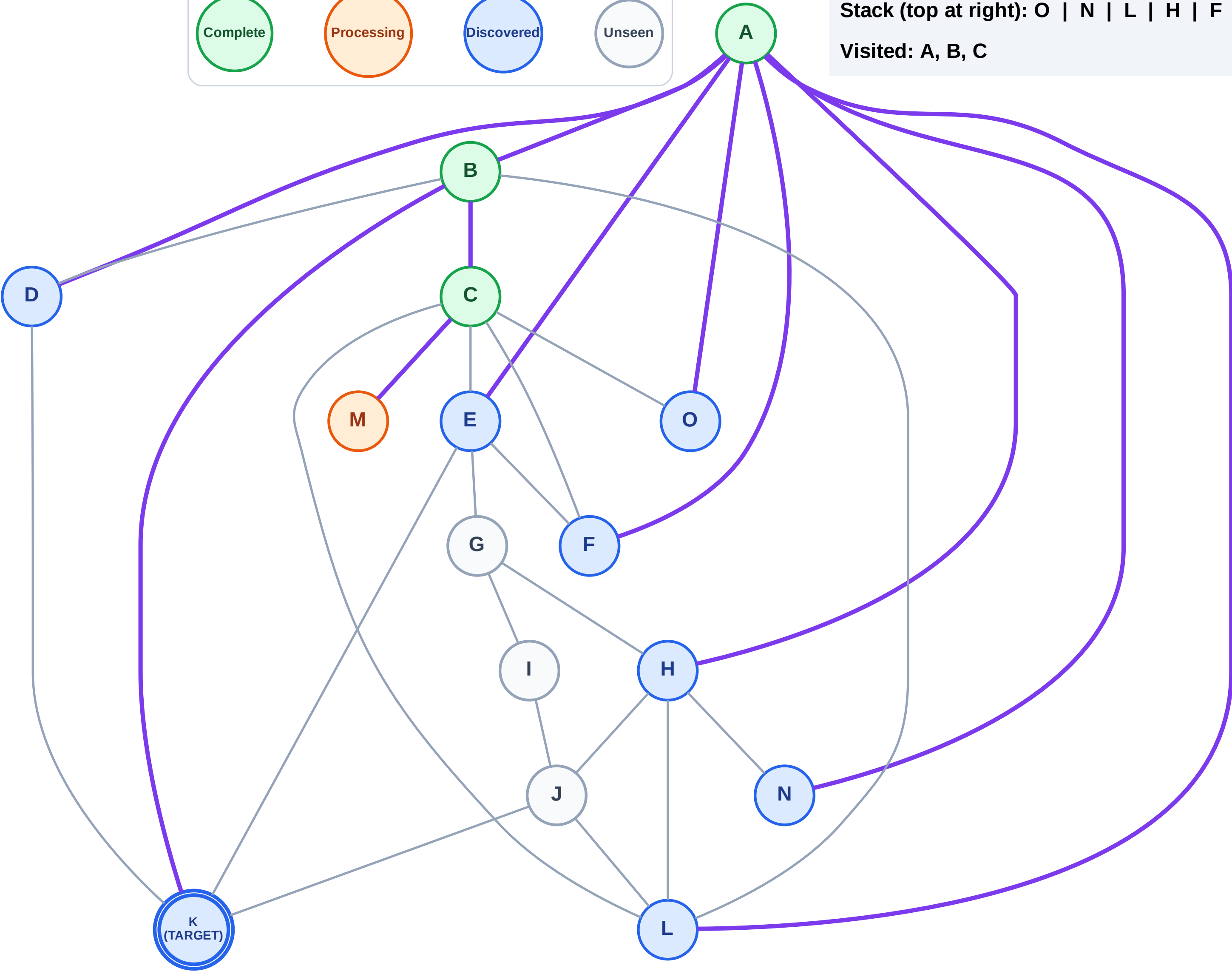
Processing

Discovered

Unseen

Stack (top at right): O | N | L | H | F | E | D | K

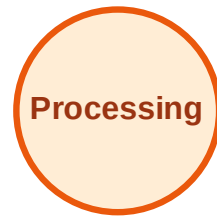
Visited: A, B, C



# DFS Traversal

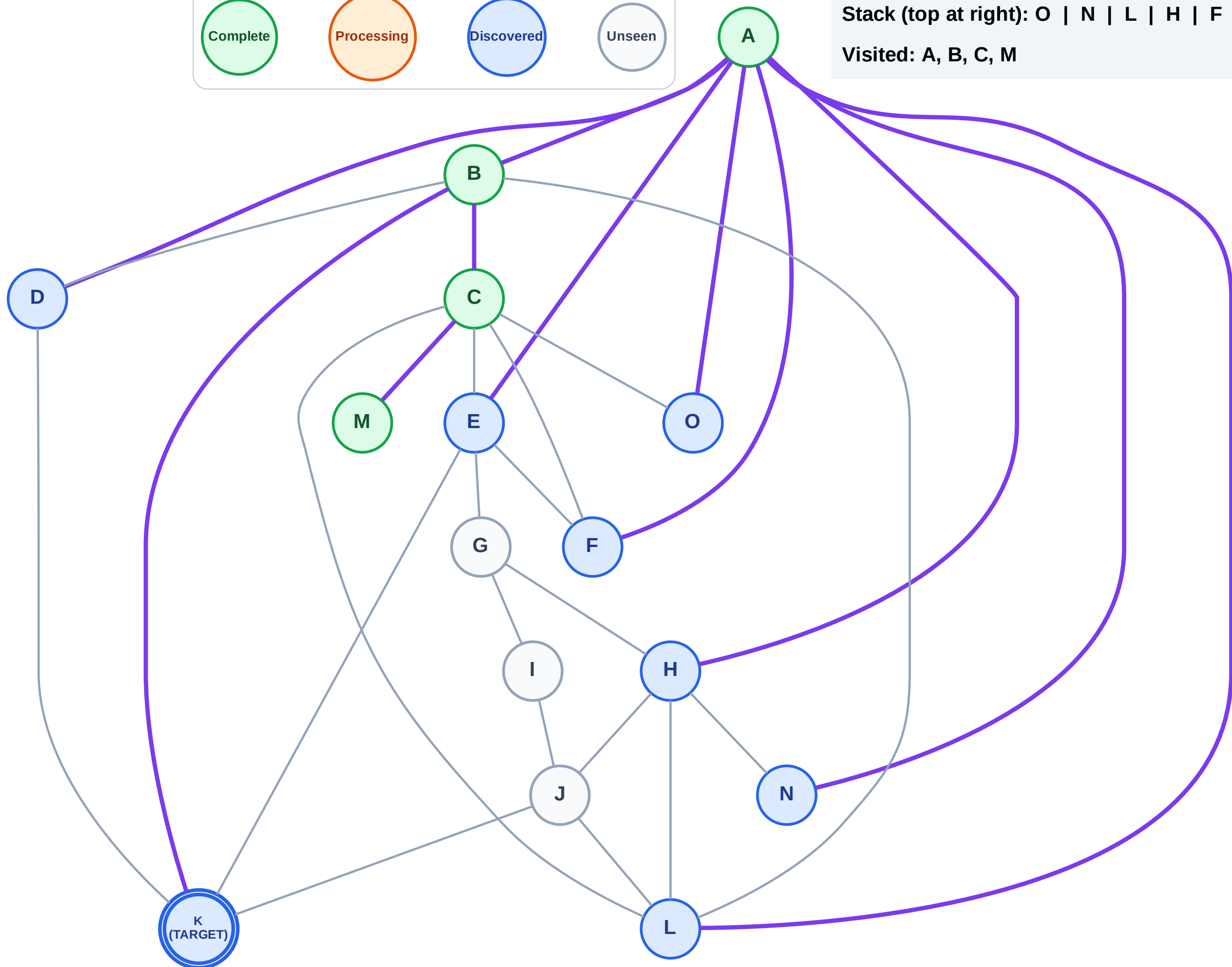
Step 9: No new neighbors from M

## Legend



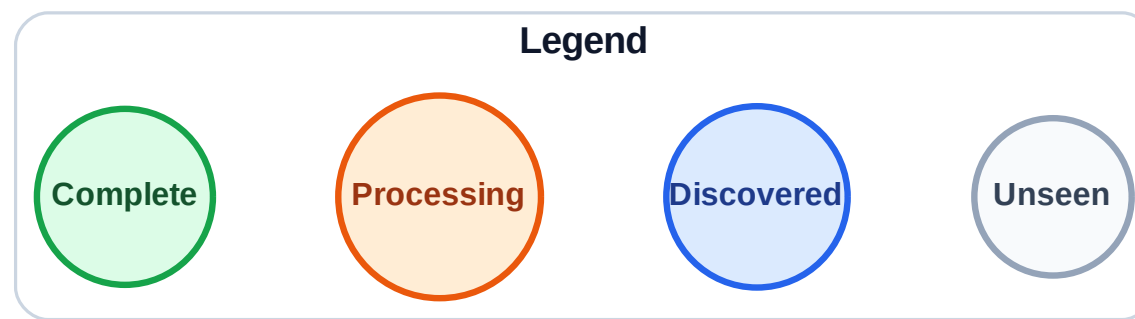
Stack (top at right): O | N | L | H | F | E | D | K

Visited: A, B, C, M



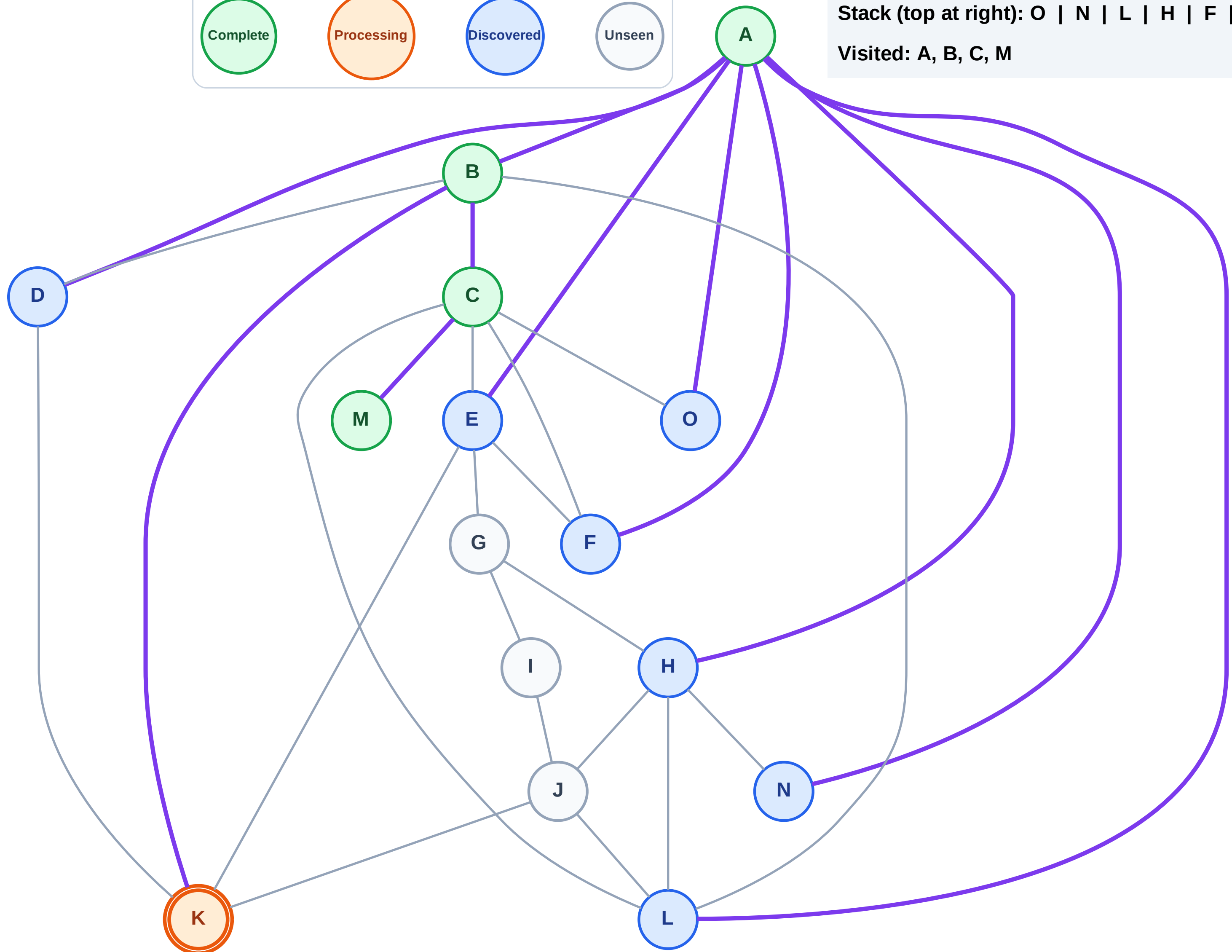
# DFS Traversal

Step 10: Process node K



Stack (top at right): O | N | L | H | F | E | D

Visited: A, B, C, M



# DFS Traversal

Step 11: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | L | H | F | E | D

Visited: A, B, C, K, M

